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Chapter - 4

Rice (*Oryza sativa* L.)

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4.1. INTRODUCTION

Rice (*Oryza sativa*) is the major food crop in the world. Nearly 40% of the world population consumes rice as the major staple food. Most of the people, who depend on rice as primary food, live in the less developed countries. Archeological evidence on rice in India dates back to 1500-1000 B.C. Since the dawn of civilization, rice has served humans as a life-giving cereal in the humid regions of Asia and, to a lesser extent, in West Africa. Introduction of rice into Europe and the America has led to its increased use in human diets. There are 42 rice producing countries throughout the world but China and India are major rice production centers. Rice is grown in wide range of agro-climatic conditions ranging from mountainous (Jammu) lands to low land delta areas (Sundarban), spanning an area from 53° latitude north to 35° south of the world but about 90% of the crop is grown and consumed in Asia. Rice provides fully 60% of the food intake in Southeast Asia and about 35% in East Asia and South Asia. The highest level of per capita rice consumption (130-180 kg per year, 55-80% of total caloric source) takes place in Bangladesh, Cambodia, Indonesia, Laos, Myanmar (Burma), Thailand, and Vietnam (Kenneth and Kriemhild, 2000). In many cultures of the world rice is the central part of people's life and culture. Rice is an excellent food and is an excellent source of carbohydrates and energy.

During last 50 years, world rice area has increased by 1.37 times from 115.50 to 159 m ha but production has increased three times from 216 to 685 mt and productivity has increased 2.3 times from 1.87 t/ha to 4.30 t/ha (Fig. 4.1). China is the world's leading rice producer with nearly 125 million tones

production. India possess largest rice area (45 m ha) producing nearly a quarter of Asia's production occupying second position after China (Moya *et al.*, 2004). In India, Rice production and productivity showed a steady increase from the first five year plan to the tenth five year plan (Fig. 4.2). Rice production has been increased during last six decades by nearly 481% or 4.8 times from 20.58 mt in 1950-51 to nearly 99.15 mt during 2008-09, whereas the average rice productivity has been increased 3.3 folds from 668 kg/ha in 1950-51 to 2186 kg/ha in 2008-09 (Table 4.1).

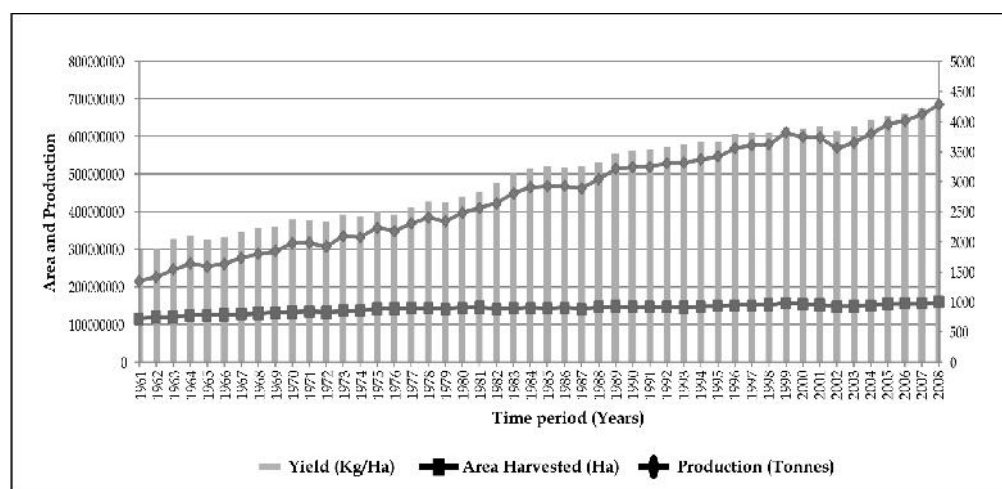


Fig. 4.1. World paddy area, production and productivity during last 48 years (Source: Anonymous, 2009a)

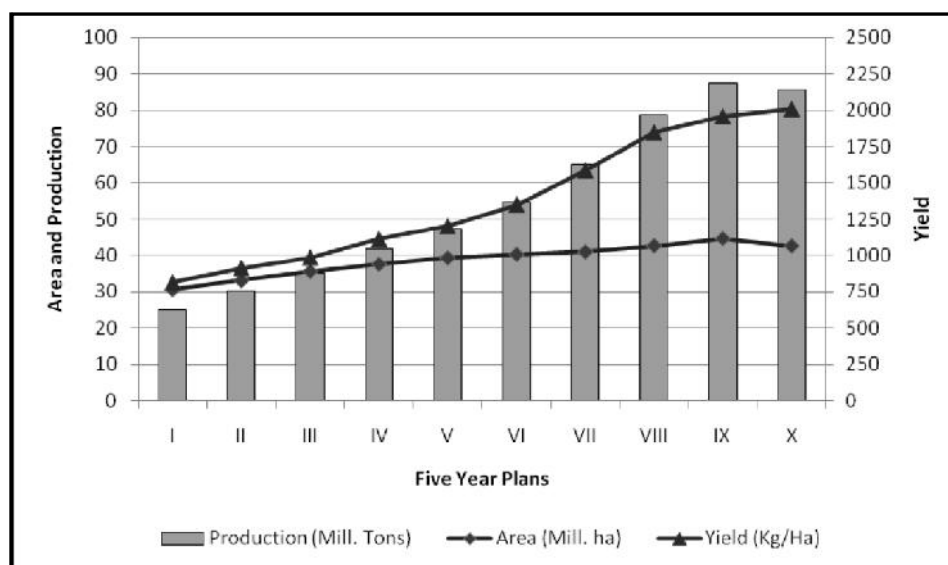


Fig. 4.2. Rice area, production and productivity in India during five year plans (Source: Anonymous, 2009a)

Table 4.1. All India rice area, production and productivity from 1950-51 to 2008-09 (Anonymous, 2009b)

Year	Area (mh)	Production (mt)	Yield (kg/ha)	Year	Area (mh)	Production (mt)	Yield (kg/ha)
1950-51	30.81	20.58	668	1980-81	40.15	53.63	1336
1951-52	29.83	21.30	714	1981-82	40.71	53.25	1308
1952-53	29.97	22.90	764	1982-83	38.26	47.12	1231
1953-54	31.29	28.21	902	1983-84	41.24	60.10	1457
1954-55	30.77	25.22	820	1984-85	41.16	58.34	1417
1955-56	31.52	27.56	874	1985-86	41.14	63.83	1552
1956-57	32.28	29.04	900	1986-87	41.17	60.56	1471
1957-58	32.30	25.53	790	1987-88	38.81	56.86	1465
1958-59	33.17	30.85	930	1988-89	41.73	70.49	1689
1959-60	33.82	31.68	937	1989-90	42.17	73.57	1745
1960-61	34.13	34.58	1013	1990-91	42.69	74.29	1740
1961-62	34.69	35.66	1028	1991-92	42.65	74.68	1751
1962-63	35.69	33.21	931	1992-93	41.78	72.86	1744
1963-64	35.81	37.00	1033	1993-94	42.54	80.30	1888
1964-65	36.46	39.31	1078	1994-95	42.81	81.81	1911
1965-66	35.47	30.59	862	1995-96	42.84	76.98	1797
1966-67	35.25	30.44	863	1996-97	43.43	81.74	1882
1967-68	36.44	37.61	1032	1997-98	43.45	82.53	1900
1968-69	36.97	39.76	1076	1998-99	44.80	86.08	1921
1969-70	37.68	40.43	1073	1999-00	45.16	89.68	1986
1970-71	37.59	42.22	1123	2000-01	44.71	84.98	1901
1971-72	37.76	43.07	1141	2001-02	44.90	93.34	2079
1972-73	36.69	39.24	1070	2002-03	41.18	71.82	1744
1973-74	38.29	44.05	1151	2003-04	42.59	88.53	2077
1974-75	37.89	39.58	1045	2004-05	41.91	83.13	1984
1975-76	39.48	48.74	1235	2005-06	43.66	91.79	2102
1976-77	38.51	41.92	1089	2006-07	43.81	93.35	2131
1977-78	40.28	52.67	1308	2007-08	43.91	96.69	2202
1978-79	40.48	53.77	1328	2008-09*	45.35	99.15	2186
1979-80	39.42	42.33	1074	2009-10*	-	99.18	-

*Advanced estimates

Note: The yield rates given above have been worked out on the basis of production & area figures taken in '000 units.

To feed the estimated world population of 9.1 billion by 2050 the overall food production has to be increased by 70% of the existing. The annual cereal

production will have to grow by almost a billion tonnes. Crop yields will grow but at a slower rate than in the past. Although rice commands a higher price than wheat on the international market, less than 5% of the world's rice enters that market, contrasted with about 16% of the wheat. On the basis of mean grain yield, rice crops produce more food energy and protein supply per hectare than wheat and maize. Hence, rice can support more people per unit of land than the two other staples (Lu and Chang, 1980). It is, therefore, not surprising to find a close relationship in human history between an expansion in rice cultivation and a rapid rise in population growth (Chang, 1987).

As a human food, rice continues to gain popularity in many parts of the world where other coarse cereals, such as maize, sorghum and millet, or tubers and roots like potatoes, yams, and cassava have traditionally dominated. For example, of all the world's regions, Africa has had the sharpest rise in rice consumption during the last few decades. Rice is unquestionably a superior source of energy among the cereals. The protein quality of rice (66%) ranks only below that of oats (68%) and surpasses that of whole wheat (53%) and of corn (49%). Milling of brown rice into white rice results in a nearly 50% loss of the vitamin-B-complex and iron and washing milled rice prior to cooking further reduces the water-soluble vitamin content. However, the amino acids, especially lysine, are less affected by the milling process (Kik, 1957; Mickus and Luh 1980; Juliano 1985a).

Rice, which is low in sodium and fat and is free of cholesterol, serves as an aid in preventing hypertension. It is also free from allergens and now widely used in baby foods (James and McCaskill, 1983). Because rice flour is nearly pure starch and free from allergens, it is the main component of face powders and infant formulas. Its low fibre content has led to an increased use of rice powder in polishing camera lenses and expensive jewelry. Rice starch can also serve as a substitute for glucose in oral rehydration solution for infants suffering from diarrhoea (Juliano, 1985b). The coarse and silica-rich rice hull is finding new use in construction materials. Rice straw is used less in rope and paper making industries than before, but except for modern varieties, it still serves as an important cattle feed throughout Asia.

In industrial usage, rice is also gaining importance in the making of infant foods, snack foods, breakfast cereals, beer, fermented products, and rice bran oil, and rice wine remains a major alcoholic beverage in East Asia.

On the research front, rewards can be gained by breaking the yield ceiling, making pest resistance more durable, and improving the tolerance to environmental stresses. Biotechnology will serve as a powerful force in broadening the use of exotic germplasm in *Oryza* and related genera (Chang and Vaughan, 1991). We also need the inspired and concerted teamwork of those various sectors of society that, during the 1960s and 1970s, made the "Green Revolution" an unprecedented event in the history of agriculture.

4.2. ORIGIN AND CLASSIFICATION

The primary centre of origin of the cultivated rice according to archeological and historical evidence is South-east Asia for *O. sativa* and Africa for *O. glaberrima*. The common ancestor to all rice is considered to be *O. perennis* and *O. sativa* is derived from perennial *O. rufipogon* and annual *O. nivara*, whereas *O. glaberrima* is derived from perennial *O. longistaminata* and annual *O. barthii*. The schematic representation of evolutionary pathways is shown in Fig. 4.3.

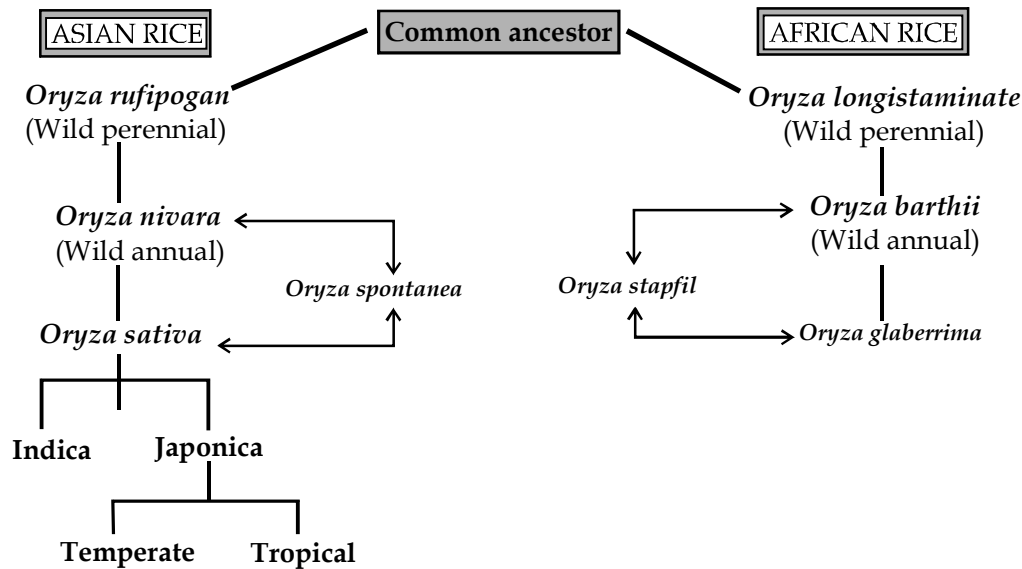


Fig. 4.3. Schematic representation of evolutionary pathways of Asian and African cultivated rice

The cultivated species *O. sativa* has evolved into three geographical sub-species due to adaptation under diverse climatic conditions for a long period, they are i) *O. sativa* sub sp. *indica*, ii) *O. sativa* sub sp. *japonica* and iii) *O. sativa* sub sp. *javanica*. Marked sterility barrier occur between these three sub-species. It ranges up to 80% in *indica* × *japonica* but to a less extent between *indica* and *javanica* cross.

The *japonica* subspecies are characterized by short, erect plant with compact panicles, pubescent spikelets and oval to round grains. These varieties are found mainly in Japan. The *javanica* subspecies are characterized by very tall, erect plants with long panicles and awned spikelets and are mainly grown in Indonesia. The *indica* sub species contains tall, spreading plant types with lot of variation in the morphological characters and are cultivated all over the tropical and sub-tropical Asia. The major differences between these three sub species are given in Table 4.2.

Table 4.2. Distinguishable characteristic different subspecies of three sub species of *Oryza sativa*

Character	Sub-species		
	Indica	Japonica	Javanica
Leaves	Broad, light green	Narrow, dark green	Broad, stiff, light green
Tillering	Profuse	Moderate	Low
Height	Tall	Short	Tallest
Hairs on lemma and palea	Thin short hairs on lemma	Dense long hairs on lemma and palea	Long hairs on lemma and palea
Awns	Mostly awnless	Awnless to long awned	Awnless or long awned
Grains	Slender some what flat grains	Short roundish grains	Broad thick grains
Grain shattering	Fast shattering	Low shattering	Low shattering
Tissues	Soft plant tissues	Hard plant tissues	Hard plant tissues
Photosensitivity	Varying sensitivity to photoperiod	Varying sensitivity to photoperiod	Low sensitivity to photoperiod

4.3. BOTANY

4.3.1. Taxonomy

Rice belongs to the genus *Oryza*, family: Poaceae (Gramineae), tribe: Oryzeae. The genus *Oryza* is distributed throughout the tropics and subtropics of the world. It has 25 species out of which 23 species are wild and only two species viz. *Oryza sativa* and *Oryza glaberrima* are cultivated species. *Oryza sativa* is mainly cultivated in the Asiatic region where as *O. glaberrima* in Africa. The genus contains both diploid ($2n = 24$) as well as tetraploid ($2n = 48$) species. Based on genome analysis all the 25 species were grouped into nine distinct genomes, viz. A, B, C, D, E, F, G, H and J and were classified in five different complexes as given in Table 4.3.

Table 4.3. Species complexes of different *Oryza* species and their geographical distribution

Sl. No	Species complex	Chromosome Number (2n)	Genome	Geographical distribution
I	Sativa Complex			
1	<i>O. sativa</i> L.	24	AA	South & Southeast Asia
2	<i>O. nivara</i> Sharma et Shastry	24	AA	Tropical Asia
3	<i>O. rufipogon</i> Griff.	24	AA	Tropical Asia
4	<i>O. meridionalis</i> Ng	24	AA	Tropical Australia
5	<i>O. glumaepetula</i> Steud.	24	AA	Tropical America
6	<i>O. glaberrima</i> Steud.	24	AA	Tropical West Africa
7	<i>O. barthii</i> A.Chev. et Roehr.	24	AA	West Africa
8	<i>O. longistaminata</i> A.Chev. et Roehr.	24	AA	Africa
II	Officinalis / Latifolia Complex			
9	<i>O. punctata</i> Kotschy ex Steud.	24	BB	Africa
10	<i>O. rhizomatis</i> Vaughan	24	CC	Sri Lanka
11	<i>O. minuta</i> J.S.Pesl. ex C.B. Presl.	48	BBCC	Philippines, New Guinea
12	<i>O. malamphuzensis</i> Krishn. Et Chandr.	48	BBCC	South India (Kerala)
13	<i>O. officinalis</i> Wall.ex Watt	24	CC	Asia, New Guinea
14	<i>O. eichingeri</i> A. Peter	24	CC	East Africa & Sri Lanka
15	<i>O. latifolia</i> Desv.	48	CCDD	Central & South America
16	<i>O. alta</i> Swallen	48	CCDD	Central & South America
17	<i>O. grandiglumis</i> (Doell) Prod.	48	CCDD	South America
18	<i>O. australiensis</i> Domin.	24	EE	Northern Australia
19	<i>O. schweinfurthiana</i> Prod.	48	BBCC	Tropical Africa
III	Meyeriana complex			
20	<i>O. granulate</i> Nees et Arn.ex Watt	24	GG	South & Southeast Asia
21	<i>O. meyeriana</i> (Zoll et Mor.ex Steud) Baill.	24	GG	Southeast Asia
IV	Ridleyi complex			
22	<i>O. longiglumis</i> Jansen	48	HHJJ	Indonesia, New guinea
23	<i>O. ridleyi</i> Hook f.	48	HHJJ	Southeast Asia
V	Unclassified			
24	<i>O. brachyantha</i> A.Chev. et Roehr	24	FF	West & Central Africa
25	<i>O. schlechteri</i>	48	HHKK	Indonesia, New Guinea

4.3.2. Morphology

Cultivated rice is an annual crop but some of its wild relatives are perennial in nature. The rice plant, typically a grass, consists of root, stem (culm), tillers, leaves and panicles at the terminal part of productive tillers. The plant height varies by variety and environmental conditions, ranging from 0.4 m in dwarf varieties to 5 m in some deep water floating rice.

4.3.2.1. Root

All the members of gramineae family are monocots with adventitious root system. The primary root which results due to the prolongation of the radical is in turn supported by several adventitious roots. Generally taller varieties have a greater penetration of roots than shorter varieties. Similarly varieties with good tillering habit have well developed root system. The direct seeded crop develops deeper but poorly developed root system, whereas the transplanted crop has a shallow but well developed root system. In case of deep water rice varieties the nodal roots develop even on higher nodes as the plant elongates.

4.3.2.2. Stem

The *Oryzae* species stem like other grass plants consists of both underground rhizome and aerial pseudostem. This condition is more visible in the wild rice with perennial habit. Where as in the cultivated species there is no well differentiated rhizome. The stem above the ground level consists of solid nodes and hollow internodes and is commonly called as *culm* (Fig. 4.4). Based upon the habitat and species the elongation of internodes varies. The rapid elongation is observed mostly after the emergence of panicle from the flag leaf. The lowermost buds on the crowded nodes just at the ground level or below that develop into tillers. The tillering habit depends on varieties, spacing, fertilizers, and cultural operations. Tillering starts at 10 days after transplanting and reaches maximum between 40 and 50 days after transplanting (Arunachalam, 1984).

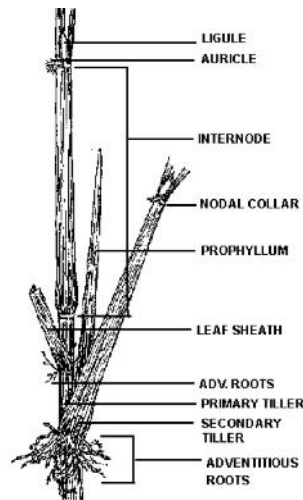


Fig. 4.4. Morphology of rice.

4.3.2.3. Leaf

The leaves arise from the nodes on the culm in two ranks. Each leaf is composed of a leaf sheath and lamina (Fig. 3.4). The junction of leaf sheath and lamina is called as *junctura/collar*, which is a triangular structure. At the base of the juncture a membranous ligule and a pair of sickle shaped auricles are present on each side of the lamina. The *ligule* is oblong or blunt and 1-5 mm long in most species. The swollen base of the sheath, above the point of insertion on the culm, is called *pulvinus*. The main culm bears large number of leaves compared to tillers. The number of leaves on tillers decreases with the rise in tillering order. The last leaf on the culm is called flag leaf/ boot leaf and its lamina is shorter and wider than that of other leaves the angle of flag leaf is oriented more vertically than the preceding leaves. The tillers arise from auxillary buds and are enclosed by special leaves called *prophyllus*.

4.3.2.4. Inflorescence

The inflorescence in rice is a terminal panicle (Fig. 4.5). The stalk of the panicle is called *peduncle* and it is the last internode of the culm. The junction of the peduncle with the main axis of panicle is marked by a ciliate ring. In rice the exposure of portion of peduncle from the juncture of boot/ flag leaf to ciliate ring is called *panicle exertion* and the degree of exertion varies between varieties. The main axis of the panicle called primary axis bears a number of secondary *rachii*. The secondary rachis further branches into tertiary rachii. The tertiary rachis in turn produces alternately arranged rachillae, which bears spikelets on their tips. The primary rachis contains hollow internodes but secondary and tertiary rachii are not hollow. The shape of the inflorescence in most rice species is oblong or fusiform.

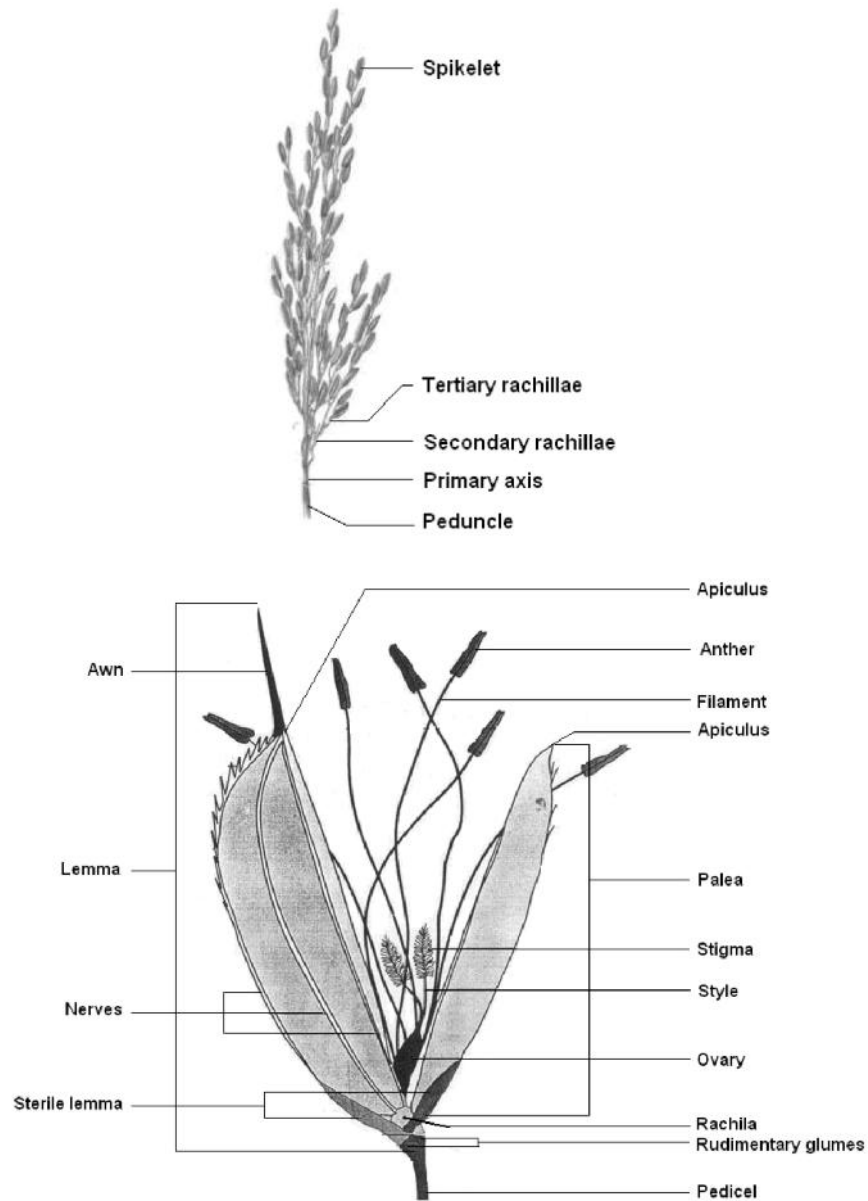


Fig. 4.5. Inflorescence, floral parts and grain of rice

4.3.2.5. Flower

The spikelet consists of two empty glumes (sterile glumes) and two fertile glumes, outer lemma and inner palea. Two transparent, fleshy structures situated at the base of spikelet are called *lodicules*, which represents the reduced perianth. Inside lemma and palea, six stamens of 2.0-2.5 mm long, arranged in two whorls of three each and a purple shaded pistil containing single ovule

with short slender style and bifurcated, plumose stigma are present. The empty glumes are generally single nerved where as lemma consists of five veins and palea. The middle vein of lemma in some varieties is elongated into an *awn* (Fig. 3.5).

4.3.2.6. Grain

The rice grain is typically a fruit called *caryopsis* and is oblong in shape. The outside hull consisting of lemma and palea encloses the caryopsis (brown rice). Brown rice consists of mainly outer pericarp followed by seed coat, nucellus, a small embryo towards ventral side and a large endosperm (Fig. 3.5). In *indica* rice lemma, palea and rachilla constitute the hull where as in japonica rice the hull is made up of mainly rudimentary glumes and sometimes a portion of pedicel. A single grain weighs about 10-40 mg. The hull constitutes nearly 20% of the total grain weight. The grain length and width varies among varieties.

4.3.2.7. Grain Quality

Grain quality is mainly a varietal character but also depends to some extent on environment, processing and handling systems. The removal of hull covering the caryopsis (brown rice) is called *dehusking*. The percentage of dehusked kernel to that of paddy is called *hulling percentage*. The removal of outer bran from the brown rice using polisher is called milling. Rice bran includes outer pericarp, seed coat, nucellus and aleurone layer of endosperm of brown rice. Percentage of polished grain to that of paddy is called milling percentage. Thus for a given sample of paddy about 70% of milled rice is obtained after the removal of hull (20%) and bran (8-10%). Even though bran consists of valuable nutrients its removal is necessary for longer storage of grain since the fat in the bran develops rancidity up on storage. The percentage of 3/4th to whole polished grains to that of paddy is called *head rice recovery*. Although head rice recovery is a inherited trait it is also dependent on temperature and humidity during ripening, grain size, shape, hardness, abdominal white, storage conditions etc. Varieties possessing medium slender, long slender and translucent grains give high head rice yields. Parboiling increases the milling percentage. Varieties with high protein content and high gelatinization temperature types are less prone to cracking.

Factors Affecting Broken in Rice

- Imbalanced nutrient application/untimely application leads to improper grain filling resulting in increased broken.
- Avoid application of nitrogen at flowering/ boot leaf stage which increases the endosperm protein causing more broken during milling.

- The produce from the saline soils may contain shriveled and ill-filled grains resulting in poor milling recovery.
- Water stress during reproductive phase (panicle initiation to physiological maturity) may lead to increased broken.
- Ensure effective control of BPH especially after flowering to reduce broken in milling.
- Protect the crop from diseases such as Sheath Blight, Sheath rot and RTV as the produce from the affected crop will be of inferior quality.
- Harvest the crop at right stage to ensure grain quality (i.e. when 80% of the panicles turned yellow and 90% of the panicles turned straw colour). The grains at the base of the panicles should be at hard dough stage.
- Over ripened and under ripened produce also leads to breakage.
- Allow the sheaves to dry in the field for 2 or 3 days to reduce the moisture content
- Whenever feasible harvest, thresh and clean the produce simultaneously using combine harvesters to get good produce.
- Avoid varietal admixtures to overcome poor milling recovery and to realize quality rice.

The appearance of milled rice is important to the consumer, which in turn is important to producer and miller. Thus, grain size and shape are the first criteria for breeding the quality rice for commercial purpose (Adair *et al.*, 1966). The International Rice Research Institute, Philippines, has given a classification for size and shape based on brown rice length and length-width ratio respectively (Table 4.4).

Table 4.4. Rice grain size and shape variation as per IRRI (2002) classification

Grain size			Grain shape		
Scale	Length	Value (mm)	Scale	Length	Value in mm
1	Extra long	> 7.5	1	Slender	> 3.0
3	Long	6.6-7.5	3	Medium	2.1-3.0
5	Medium	5.51-6.6	5	Bold	1.1-2.0
7	Short	d" 5.5	9	Round	< 1.1

The Indian government appointed Ramaiah committee in 1965 to recommend a uniform standard of grain classification (Table 4.5). The grain classification in India is being followed according to that committee recommendation.

Table 4.5. Rice grain shape based on length and length/breadth ration followed in Indian sub continent (Anonymous, 1985)

Grain Shape	Character
Long slender (LS)	Length 6 mm and above, length/breadth ratio 3 and above
Short slender (SS)	Length less than 6 mm, length / breadth ratio 3 and above
Medium slender (MS)	Length less than 6 mm, length / breadth ratio 2.5 to 3
Long Bold (LB)	Length 6 mm and above, length / breadth ratio less than 3
Short Bold (SB)	Length less than 6 mm, length / breadth ratio less than 2.5

For commercial purpose the Food Corporation of India has classified rice varieties into three groups in 1978 based on their length / breadth ratio later in 1998 they have reclassified the three groups into two groups as follows:

Length / breadth ratio	1978 Classification	1998 Classification
Less than 2.5	Common	Common grade
2.5 to 3.0	Fine	Grade A
Above 3.0	Super fine	

The Food Corporation of India while purchasing the paddy grains imposes some quality norms famously known as Fair Average Quality (FAQ) norms both for Grade A and common varieties (Table 4.6). Only the produce meeting these norms will be procured by FCI.

Table 4.6. Fair Average Quality norms of Paddy (Anonymous, 2009c)

Refraction	Maximum limits (%)
Foreign matter: Inorganic	1.0
Organic	1.0
Damaged, Discoloured Sprouted and weevilled grains	4.0
Immature, Shrivelled and shrunken grains	3.0
Admixture of lower class	7.0
Moisture	17.0

Cooking and eating characteristics are largely determined by the starch properties which accounts for 90% of the milled rice. *Gelatinization temperature*, amylase content and gel consistency are important starch properties which influence cooking quality. The gelatinization temperature of rice varieties may be classified as low (55-69°C), intermediate (70-74°C) and high (>74°C). In rice breeding programs Alkali Spreading Value technique is extensively used for the estimation of Gelatinization temperature. Rice with low gelatinization temperature disintegrate completely in alkali solution (1.7% KOH), whereas rice with intermediate gelatinization temperature disintegrates partially. Rice with high gelatinization temperature is largely unaffected in the alkali solution.

Gelatinization temperature and cooking time of milled rice are positively correlated (Juliano *et al.*, 1964). *Amylose* content is considered to be the single most important character that influences the rice cooking and processing behavior (Webb, 1985). Rice varieties are grouped into five categories based on their amylose content. They are waxy (0-2%), very low (3-9%), low (10-19%), intermediate (20-25%) and high (>25%). The waxy (glutinous) rice does not expand in volume and are glossy and sticky and remain firm when cooked whereas high amylose rice show high volume expansion and a high degree of flakiness.

Nutritionally rice is rich in carbohydrates. The protein content of the grain on an average is 7% in milled rice and 8% in brown rice. Rice contributes to 40-80% of calories and 40 percent of protein in the Asian diet. The amino acid balance of rice protein is also good. Brown rice contains important B-complex vitamins and minerals viz. thiamines, riboflavin, niacin, calcium, iron etc.

4.3.3. Reproductive Biology

Rice is basically an *autogamous* plant propagating mainly through seeds produced by self pollination. Normally *anthesis* starts shortly after the panicle emerges from the flag leaf. The opening of flowers within a panicle starts from top to bottom and it takes nearly one week for the anthesis to complete. The spikelets generally opens from morning 9-10 am but it varies with the prevailing weather conditions particularly temperature and humidity. After the anthesis the protruded anthers will wither away. Each anther consists of more than 1000 pollen grains, which disperses immediately after the spikelet opening and germinates on the surface of stigma in about 3 minutes after pollination. Only one pollen tube reaches the ovule in about an hour after pollination, to initiate *double fertilization* and *triple fusion*. The maturation of anther and ovule occurs simultaneously without any time gap within a spikelet. Germinability of pollen lasts only for few minutes under favourable temperature and moisture conditions whereas ovules can remain viable for several days after maturation. The pollen viability of cultivated species lasts only for 3-5 minutes while that of wild species it will be up to 9 minutes (Oka and Morishima, 1967; Koga *et al.*, 1971). Even though rice is basically a self-pollinated crop, natural outcrossing can occur up to 0.5 to 4% (Oka, 1988). Most of the wild species have longer stigma, which protrudes well outside the spikelet facilitating increased percentage of out crossing (Virmani and Edwards, 1983). Among different ecotypes of *O. sativa*, *indica* have higher percentage of out crossing than *japonica* (Oka, 1988).

Although *O. sativa* is an annual crop propagated by seed, it can be maintained vegetatively as a perennial crop. The perennial character *O. sativa* might have been inherited from its ancestor *O. rufipogon*, a perennial species (Morishima *et al.*, 1963). Under natural conditions after harvesting, the tiller

buds on the basal nodes starts growing again. These new tillers are called “*ratoon*” and can be grown best under long day conditions.

The production of viable hybrids between *O. sativa* and its distant relatives is difficult to achieve mainly due to the reproductive barriers existing between them. The post fertilization barriers are of mainly four types, viz. F_1 inviability (early embryo abortion), F_1 weakness, F_1 sterility and hybrid breakdown (Oka, 1988).

4.4. CYTOLOGY AND GENETICS

Kuwada (1910) first reported the presence of 24 chromosomes in diploid condition and 12 in haploid state in *Oryza sativa*. The 24 somatic chromosomes were classified into 5 large, 4 medium and 3 small pairs (Fig. 4.6) based on their morphology (Sethi, 1937). The longest chromosome measured 2.1μ and the smallest 1.4μ . The wild species, *O. latifolia*, *O. minuta*, *O. alta*, *O. eichingeri*, and *O. coarctata* have $2n = 48$ chromosomes and are tetraploids. So the crosses between *O. sativa* and tetraploid species are sterile in nature. Even the crosses between the two cultivated species *O. sativa* and *O. glaberrima* are pollen sterile.

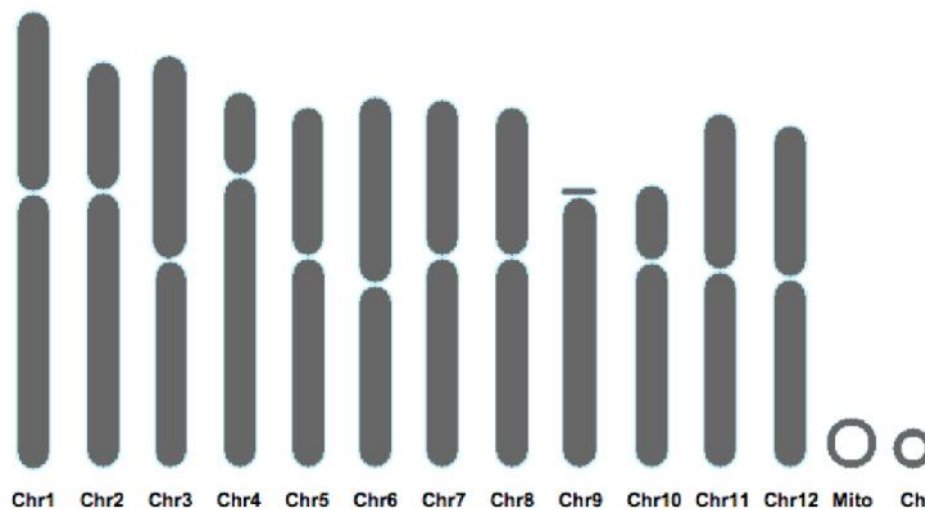


Fig. 4.6. Rice Chromosomes

The genetic nature of some of the important characters is given below which will be useful for the designing the breeding programme (Table 4.7).

Table 4.7. Genetic nature of some important characters of rice

Character	Dominant	Recessive
Plant height	Tall	Short
Hairyness	Hairy	Non-hairy
Lodging	Lodging	Non-lodging
Panicle exertion	Exertion	Non-exertion
Awn	Awned	Awnless
Hairyness on glumes	Pubescence	Smoothness
Grain length	Short grain	Long grain
Starch condition	Starchy grain	Glutinous

4.5. GENETIC DIVERSITY

The genus *Oryza* has 25 species out of which only two are cultivated species. In Indian sub continent *Oryza sativa* is the cultivated species. India is rich in rice diversity by virtue of being a part of primary centre of origin on one hand and having many secondary centres on the other hand. There are thousands of land races and hundreds of improved cultivars adapted to different agro-ecological conditions are present in India. According to an estimate there are 75000 to 100000 land races are present in India (Randhawa *et.al.*, 2006) and in last four decades (1965-2007) more than 836 high yielding rice varieties were released (Shobha Rani *et al.*, 2008). India is also known for quality rice like Basmati and other fine grain aromatic types and many cultivars with medicinal value viz. Gauthuhan, Njavara, Alcha, Laicha etc. More than 90000 accessions have been imported and made available to researchers and farmers in India (Singh *et al.*, 2001). Introductions from IRRI during 1960s have led to green revolution. Characterization of germplasm based on identifiable stable morpho-physiological traits has enabled their use for economic importance. Characterization of wild rice species has revealed their importance as potential donors for various biotic and abiotic stresses. A number of elite lines and cultivars characterized with resistance to biotic and abiotic stresses and quality characters were developed through conventional breeding using the available genetic variability in the germplasm of *O. sativa* to the maximum extent. However genetic variability for different traits like resistance to stem borer, sheath blight, tungro virus etc. is limited in the cultivated germplasm. Therefore, for further improvement the cultivated rice gene pool has been widened by introgression of genes from wild species for tolerance/resistance to major biotic and abiotic stresses as well as for quality characters using different newly developed techniques which were discussed in the later part of this chapter.

4.6. BREEDING FOR IMPROVEMENT OF RICE

Rice breeding in India started in 1911 in undivided Bengal in the appointment of Dr. Hector as economic botanist, with its head quarters at Dhaka. The 71 years from 1911-1983 falls into three distinct periods 1) of pure line selection, 2) of interracial hybridization between indica and japonicas and 3) of inter-racial hybridization with semi dwarf specially Taiwanese indices (for detail see Section 1.2.1 of Chapter 1).

The main aspect in breeding is creation of variation and selection of suitable lines. For the development of targeted varieties with specific characters, one of the parents should possess the necessary characters for obtaining a population with desired traits. Such a parent is called donor. The large variability in rice genetic resources has led to identification of several donor sources suited to specific and changing agro-climatic conditions, having resistance/tolerance to biotic/abiotic stresses and with yield supporting physiological traits.

4.6.1. Breeding Objectives

The broad breeding objectives of rice in India/Asia are:

4.6.1.1. Increasing the Yield Potential

The productivity of rice is very low in tropical countries like India compared to temperate countries like China, Japan, and Taiwan etc. Thus productivity of rice has to be increased through different breeding approaches. Yield is a complex character which may be influenced by many physiological processes within the plant. It is also influenced by environmental parameters of the rice growing area. To increase the yield, New Plant Type (NPT) concept may be followed.

4.6.1.2. Biotic Stress Breeding

The biotic stresses mainly the disease and insect pests are one of the major areas which reduce the yield of crop if it not controlled at proper stage of attack or infestation. Some time the disease may become epidemic leading famine in the affected areas, for example Bengal famine in India during 1943. Some of the biotic stresses (diseases and insect pests) are being enlisted below and the donor for resistance listed in Table 4.8.

- Bacterial blight (*Xanthomonas campestris*)
- Blast (*Pyricularia oryzae*)
- Stem rot (*Leptosphaeria salvinii*, *Helmenthorium sigmoideum*)
- Brown spot (*Helmenthorium oryzae*)
- Tungro (Viral disease)
- Brown plant hopper (*Nilaparvata lugens*)

- Gall midge (*Oreseolia oryzae*)
- Stem borers (*Chilo suppressalis*, *Tryporyza innotata*, *Tryporyza incertulas*, *Sesamia inferens*)

Table 4.8. Important donors for resistance/tolerance to biotic stresses

Trait	Important donors
Insect-pests	
Yellow Stem Borer	ADT2, ARC10257, ARC10598, ARC5500, ASD10, Ptb15, Ptb18, SLO17, TKM6, CB I, CB II, ARC 6158, ARC10386, ARC10443, NCS266, NCS 336, NCS 464, W1263, Manoharsali
Gall midge	ARC5984, Bengle, Bhumansam, Eswarakora, Velluthachira, Leaung152, NHTA8, Siam29, W1263, Phalguna, Ptb18, Ptb21, OB677, ARC10660, ARC6605, ARC5959, ARC13516, ARC14787, CR94-1512-6, Shakti, Abhaya, Dukong,1, RP2333, Jhipiti, MR1523, Kranti
Brown Plant Hopper	ARC6650, ARC10550, Swarnalata, T12, Chinsaba, IR1154-243, Kaharamana, Babawee, Chemban, Co25, HR19, Mudgo, BM71, Ptb18, Ptb19, Ptb21, Ptb33, Ptb34, CR94-13, Leb Muey Nahng, Rathu Heenati, Sinna Sivappu, Suali, DV85, AS20-1, ARC7080, ARC14766, NCS91, NCS131, NCS707, <i>O. rufipogon</i> , <i>O. minuta</i> , <i>O. officinalis</i> , <i>O. latifolia</i> , <i>O. australiensis</i> , <i>O. glaberrima</i> , <i>O. eichingeri</i>
White Backed Plant Hopper	ARC10239, ARC11316, ARC6650, ARC5984, ARC6064, ARC7138, ARC7318, Asominori, ARC10340, ARC5803, Chempar, HR22, Kalubmati, Podiwi A8, B5, Giu-yi-gu, N'Diang Marie, N22, Senawee, WC1240, ADR52, Velluthachira, MO1, <i>O. glaberrima</i>
Leaf folder	ARC 10982, GEB24, Ptb33, TKM6, ARC1129, Ptb12, Gorsa, Darukasali
Green Leaf Hopper	Phankhari 203, ASD7, IR8, Ptb8, TAPL 796, Moddai Karuppan, DV85, IR28, Asmaita, Ptb2, Ptb21, ARC6606, Pe-bi-hum, ARC10313, <i>O. rufipogon</i>
Diseases	
Blast	Tetep, Tadukan, Zenith, Carreon, Co4, Raminad Str.3, Gampai, Dawn, Peta, Sigadis, Dular, Rasi, IR64, Ptb10, Ceysron, Milyang83, ARC7098, Kalamdani, Tarabali, Mugisali, Madrisali, SolIpona, Rongaahu, Gajepsali, Beganbisi2, Rongagutia, Kolimekuri, Rikhojoi2, <i>O. minuta</i> , <i>O. rufipogon</i> , <i>O. glaberrima</i>
Bacterial Leaf Blight	Tetap, Aus Boro lines, PI231129, BJ1, Karjat, Pallavi, Zenith, Malakit, Sung Song, Syntha, TKM6, Sigadis, DZZ 192, DV85, DV86, HX3, Minghui63, Lotasali, ARC5827, Dular, Pelita1, IR8, Java14, ARC18562, Asominori, IR24, XM5, XM6, Zhachanglong, Dholamula, Moinasail, Cemposalak, Chinsura Boro II, Sayaphal, Wase-Aikoku, Japorisali, Ajepsali, Jatiosali, Cas209, RP9-3, AS330, Kola ahu, <i>O. rufipogon</i> , <i>O. longistaminata</i> , <i>O. minuta</i> , <i>O. officinalis</i> , <i>O. latifolia</i> , <i>O. australiensis</i> , <i>O. brachyantha</i>
Rice Tungro Virus	Kataribhog, Latisali, ARC14766, ARC14320, ARC11554, ARC12981, ARC14766, ARC15570, BJ1, Gampai15, Ptb18, Ptb21, Pankhari203, Sigadis, Ambemohar, Habiganj, <i>O. rufipogon</i>
Rice grassy stunt virus	ARC13820, ARC13901
Sheath blight	ARC10531, ARC10532, ARC10635, ARC15362, Athebu, BCP3, Buhjan, KRC355, KRC356, Laka, Nangmons4, OS4, Phoure1,

contd.

	Ramadja, Saibham, Suduwee, T141, Ta-poo-cho-z
Brown spot	Ch 13, Ch 45, BAM 10, AC2550, Bhatta Dhan
False smut	MNP 85, ARC5378, BR16, IR24, IR29
Stem rot	Basmati 370, Bara 62
Ragged smut	Ptb21, Ptb33
Grassy stunt	<i>O. nivara</i>

4.6.1.3. Abiotic Stress Breeding

Abiotic stresses are major constraints for rice in specific areas over the globe which limits the crop production. Dudal (1976) estimated that only 10% of world's arable land may be categorized as free from stress. The rapid change in environmental conditions likely to override the adaptive potential of plants, this environmental changes mainly originated from anthropogenic activities, which has caused soil and air pollution, plant are exposed to natural climatic or edaphic stresses, such as, high irradiation, heat, chilling, freezing, drought, excess water and water-logging and nutrient imbalance. Among abiotic stresses drought is the main abiotic factor as it affects 26% of arable area. Water stress is a single most severe, limitation to the productivity of rice in the rain-fed ecosystem (Widawgky and O'Toole, 1990). Mineral toxicity/deficiencies are second in importance. Among mineral toxicity, salinity is wide spread and is estimated to affect 10% of the world land surface (Richards, 1995). Increased salinization of arable land is expected to have devastating global effects, resulting in 30% land losses within the next 25 years and up to 50% by the year 2050 (Wang *et al.*, 2003). Singh and Bandyopadhyay (1996) reported that salt affected soils occupy about 2.5×10^6 ha in India. Soil acidity is another major problem common to tropical regions, which constitutes about 3.95 billion ha of land (FAO, 1991). Acid soils caused by combination of aluminium (Al) and manganese toxicity are major constraints to soil fertility and crop productivity. Al toxicity problems are of enormous importance in production of rice, maize, and sorghum. The yield losses caused by various abiotic stresses in South and Southeast Asia have been compiled in Table 4.9 (Dey and Upadhyay, 1996). There is serious concern for food security of developing countries, which demands a conscious effort to improve production from areas commonly exposed to abiotic stresses.

Table 4.9. The fraction of world arable land affected by abiotic stresses

Abiotic stress	Fraction (%) of arable land
Drought	26
Mineral (Toxicity / deficiency)	20
Freezing	15
No stress	10

Source: Dudal, 1976

Existing tolerant germplasm may be utilized for the improvement of rice against different abiotic stresses. A list of important donors for different abiotic stresses is being given below (Table 4.10).

Table 4.10. Important donors for resistance/tolerance to abiotic stresses

Trait	Important donors
Drought	N22, MTU17, Lalnakanda41, Kalakeri, JBS508, CR143-2-2, Saria, Sathchali, Nepalikalam, Dehula, Brown gora, Dular, Kodibudama, Chanda2, DJ5, EB17, Goradhan, Gajgaur Dhani, Kinandang Patong, SLO16, Nandi, Tuljapur1, Tuljapur28, Tuljapur34, Prage147, NC487, Mettamolagolukulu, NC488, ARC10372, Janki, NC492, AS180, Hasakumra, Maibi, Koijapori, AS313/11, AS47, Bairing, Ahu joha, Dagaranga, Noga ahu, Soraituni, Lakhi, MAU Sd10, Rikuto Norin.
Low light	Ambemohar 157, Batkpahi, Gajepsali, K540, Kolapakhi, Lothabar, Malkolam, Nakarasali, Rajipsali, Z63.
Salinity	Chettiviruppu, Chootupokkali, Nonasail, Patnai 23, Pokkali, SR26B
Submergence	FR 13A, FA 43B, Chakia59, CN540, S22, Madhukar, S24, S25, S28, Boku, Boyan, Bayyahunda, HbDW8, Solpona, Sail badal, Dhola badal, Kolasali, Boga bordhan, Khajara, Dhusara, Nali Baunsagaja, Rongasali, <i>O. rufipogon</i> , Sub-1
Cold tolerance	Dunghansali, AC540, Sita, Rajai, CB1, Boro33, Dholiboro, IRGC 100081, IRGC 100114, IRGC 100028
Water logging	Jhingasail, Kalakhersail, Patnai 23, NC496, Tilakkachari
Deep water	HBj1, Nageri Baoorsa, Kekoa bao, Jalmagna, Jaladhi 1, Jaladhi 2

4.6.1.4. Quality Improvement

Grain quality is important for marketing as well as for nutritional status. Existing germplasm may be utilized for the improvement of rice grain physical quality and nutritional status. A list of important donors for quality characters is being given below (Table 4.11).

Table 4.11. Important donors for quality characters

Trait	Important donors
Grain quality	Haryana Basmati, Pakistan Basmati, Basmati 370, Improved Sabarmati, Kasturi, Pusa 33, Basmati 320, Taraori Basmati, Basmati 386
Iron	Type 3, Jalamagna, Madhukar, Kranthi, Shashi, BR 2655, Chittimutyalu, Kalanamak, ADT43, Basmati 386
Zinc	Ranbir Basmati, Vikas, Poornima, DL 183, Chittimutyalu, Kalanamak, ADT43, Basmati 386

4.6.1.5. Breeding for Physiological Attributes

Most recent concept of crop improvement is the breeding based on physiological attributes. Existing germplasm may be utilized for the

improvement of rice grain physical quality and nutritional status. A list of important donors for physiological traits is being given below (Table 3.12).

Table 4.12. Important donors for yield supporting physiological traits

Trait	Important donors
High photosynthetic efficiency	AC4491, Bam 3, Mahsuri, NS1281, Ptb10, T141, T90
Better utilization of solar energy	AC1491, Bam 3, Ptb10, T90
Low photorespiration	B76, Patnai 23
High dry matter production	Intan, Manosarovar, Pankaj, Swarnaprabha
High density grains	Badshabhog, Rounsail
High harvest index	T141, Vijaya
High fertilizer use efficiency	IR42-25323, IR29912-56-3, IR29912-56-3-1
Heavy panicle	Darinagan, Djawa serang, G1291, G1298, Ketan gebat
Prolonged grain filling	BG 380, BG90-2, Darinagan, IR65601-10-1-2, IR65740-ACI-3
Short stature	IR5, MD2, Shend Nung 889-336
Slow leaf senescence	G110, Pusa 1021, Pusa 1266, indica x japonica derivatives
Sturdy stem	Gharbaran, Pusa 44-37, Sengkeu, Sipapak, Sirah Barch
High translocation efficiency	Ptb10

Sources: Siddiq *et al.* (2005) and Shobha Rani *et al.* (2008)

Emphasis on grain quality results in unstable yields. Conversely too much emphasis on disease and insect resistance and stable yields leads to poor grain quality. Therefore, the following breeding approaches need to be emphasized for producing rice varieties with sustainable rice production-

- High yield potential with low inputs
- Good grain quality and suitable for processing
- Durable resistance to major pests and diseases of the region
- Wider range of growth duration for different seasons and purposes
- Proper level of tolerance to abiotic stresses of the region.

Since all these approaches are inter related too much emphasis on any one approach should be avoided. An integrated approach like tolerance to pests and diseases coupled with good grain quality with a yield potential of the existing high yielding varieties will serve the purpose. Based on the region specific conditions and requirements those characters are to be incorporated rather than trying to breed an all in one variety is important to observe while producing new varieties.

4.6.2. Breeding Methods

The main breeding methods are Mass selection, Pure line selection, Pedigree method, Inter varietal hybridization, Inter specific hybridization, Mutation breeding and Heterosis.

4.6.2.1. Introduction

It is most easy and first step of plant breeding. One of the important requirements of plant breeding is to have available large number of germplasm collections to serve as storehouse of genes to which plant breeder may have access. Introduced genotype may be directly release for cultivation or may be used for further breeding programmes. A number of varieties with short stature, short duration, photoperiod insensitive, fertilizer responsive, lodging tolerant and high yielding potential have been introduced from Taiwan in India. Some of the varieties introduced from IRRI, Philippines still performing well, namely IR 8, IR 36, IR 64 etc.

4.6.2.2. Mass Selection

In this method, a large number of plants with similar and desirable morphological characters are selected and their seeds are mixed together to constitute the new variety. The new variety consists of mixture of several pure-lines. This is an obsolete method and the improvement attained using this method is only temporary in nature.

4.6.2.3. Pure-Line Selection

In pure-line selection, large numbers of homozygous self pollinated plants with similar morphological characters are selected followed by individual progeny selection results in the development of pure-line varieties. Most of the earlier rice varieties were developed using this method.

4.6.2.4. Pedigree Selection

Most common method adopted in rice varietal development. Individual plants are selected from segregating F_2 generation onwards followed by selfing and selection of required characters up to F_6 generation. The individuals ancestry record called pedigree record is maintained in detail in this method which allows to track back the ancestors of newly developed variety.

4.6.2.5. Intervarietal Hybridization

The three sub species of *Oryza sativa*, indica, japonica and javanica can be easily crossable with high fertility percent. This resulted in the development of Indica-Japonica and Indica-Javanica hybrids with many useful traits and good heterosis.

4.6.2.6. Interspecific Hybridization

With the help of biotechnology tools (such as, embryo rescue, somatic and hybridization) the interspecific hybridization using wild species for agronomically important traits like yield and quality as well as resistance to biotic and abiotic stresses can be exploited to meet the present day challenges.

4.6.2.7. Mutation Breeding

Mutations can be used as a source of variation. The creation of mutations by artificial means for breeding of new varieties is called mutation breeding. The main purpose is to create variation in the available genetic material which will be used for further selection. T. 141 was developed following X-ray treatment at Orissa Agriculture University and Technology.

4.6.2.8. Heterosis

The hybrid vigour in rice, a self pollinated crop, is successfully exploited in China. The practical difficulty in large scale production of hybrid seeds was overcome by the development of cytoplasmic male sterile female lines and identification of successful restorers. The details of hybrid rice production and its importance was discussed in detail in separate chapter in this book.

4.6.3. Crossing Technique

The crossing in rice is a difficult task. Since each spikelet turns into a single grain crossing needs to be done on at least in 3-4 panicles for each set. The small size of the florets makes the task further difficult. The crossing technique consists of mainly three steps: Emasculation, Pollination and Bagging & Tagging.

4.6.3.1. Emasculation

Generally emasculation will be done either by emasculator or manually by clipping the spikelets. Even though suction emasculation through vacuum, hot water and hot air methods are there but are not popular in rice. The spikelets are clipped and emasculated in the afternoon. The spikelets which are on the second or third day of blooming are the best for this purpose. The spikelets at the top, which may be already fertilized, and the younger spikelets at the base of the panicle should be removed. Proper care should be taken while emasculating the anthers otherwise the gynesium part may be damaged leading to no seed set. Immediately after emasculation the panicles should be covered with paper bag to avoid any sort of pollination.

In rice anthesis commences shortly after emergence of panicle. Spikelets at the tip bloom first and proceed downwards. Anthesis time 8-10 am. Each spikelet remains open 30 minutes and then closes. The anther dehiscence takes place immediately after the opening of the spikelets. Receptivity remains for one day.

Emasculation is necessarily followed by controlled pollination. Emasculation is done during early morning between 6 and 8 AM in spikelets, due to open on the same day. Emasculation should be over well ahead of the time of anthesis. Crossing techniques in rice differ based on the method of emasculation. Since maximum number of spikelets open on the 3rd or 4th day of anthesis, panicles of that stage are selected for emasculation. The following methods are widely used for hybridization in rice.

i) Clipping method

In the previous day evening, top 1/3rd and bottom 1/3rd portions in the panicle of the desired female parent are clipped off by using scissors leaving the middle spikelets. With the help of scissors again, top 1/3rd portion in each spikelet is clipped-off. The six anthers present in each spikelet are removed with the help of the needle or forceps. Care must be taken during emasculation for not to damage the gynoecium. Then to prevent contamination from the foreign pollen, the emasculated spikelets are covered with a butter paper big. In the next day morning (usually at 9.00AM), the bloomed panicle from the desired male parent is taken. The top portion of the butter paper bag which was originally inserted in the emasculated female parent is now cut to expose the panicle. The male parent panicle is inserted in an inverted position into the butter paper bag and jerked in both ways in order to disperse the pollen. After ensuring the abundant disbursement of pollen, the opened butter paper bag is closed using a pin. Coloured thread may be tied at the base of the panicle to identify the crossed ones. After ensuring pollination and fertilization, the bag may be removed.

ii) Hot water method

A method of hot water emasculation is used to about the same extent as the clipping method. Panicles in 3rd (or) 4th day of blooming are chosen as female parents. An hour or so before blooming (i.e. normally at 7.00 A.M.), the panicle is selected and under developed and opened spikelets are removed. Now, the tiller is bent over (carefully to avoid breaking) and the selected panicle is immersed in hot water contained in a thermos bottle at 40-44 °C for a period of 5 to 10 minutes. This treatment causes the florets to open in a normal manner and avoids injury. Then, emasculation is done by removing the six stamens by fine forceps or needles and then dusting should be done.

iii) Dr. Ramiah method

Panicles on the 3rd or 4th day of its blooming are selected; top and lower spikelets are removed leaving only the middle. It is covered with a wet cloth and air is blown from mouth. This facilitates opening of spikelets. After 2-3 minutes, wet cloth is removed and spikelets are found to be open. Then, the six anthers are removed.

iv) Vacuum emasculation method

This works on the principle of suction pressure. The spikelets are clipped off prior to operation. The minute pipette is to be shown at the point of clipping and pollen is sucked in. Six panicles can be emasculated at a time. By hand emasculation, 100 flowers can be emasculated by a person. With the vacuum emasculator, six persons can operate and emasculate 3000 to 3600 florets/hour.

v) Cuttack Method

The technique was developed by CRRI, Cuttack. The panicle to be emasculated is inserted into hollow piece of bamboo closed at one end and plugged with cotton wool and split cork at the other end. The flowers thus enclosed will open within 5-10 minutes. The anthers are removed.

vi) Brown paper method

The panicles are enclosed in a Brown paper cover before a couple of hours of blooming. Heat develops inside due to which the anthers extrude, but do not dehisce. This happens in 15-30 minute then the anthers are easily clipped off. Stigmatic surface is then dusted with pollen grains collected from the chosen male parent. The crossed panicle is then properly tagged and protected with paper cover which is retained in a position for 7-10 days.

vii) Rhind's method

In this method hot water is kept in the flask and it is poured outside. After pouring out the water inside of the flask will be warm and humid. The panicle to be emasculated will be inserted into the flask and kept for some time. Due to high temperature and humidity the spikelets will get opened and the anthers are exposed which can be removed with the help of forceps.

4.6.3.2. Pollination

Generally pollination is done in pollination chambers where artificial heating elements arranged for hastening the anthesis. The panicles collected from the field were kept in a glass beaker containing water. Immediately after anthesis the pollen should be dusted over the emasculated spikelets. Care should be taken so that all the emasculated spikelets of female parent were properly dusted with the pollen of the male parent.

4.6.3.3. Bagging and Tagging

Immediately after dusting, the panicles are covered with butter paper bags to avoid foreign pollen contamination. The bags should not be opened till the seed sets. The crossed plants should be tagged with appropriate cross combination details for pedigree maintenance.

4.7. BIOTECHNOLOGY

Farmers around the world are gradually embracing the new technology viz., Biotechnology after the first commercialization of transgenic crops in the year 1996. In 2009 the total biotech crop area reached to 134 m ha. A record of 14 m farmers in 25 countries across the globe has planted biotech crops in 2009. At present 11 biotech crops are under cultivation in 25 countries (James, 2009) but still rice is not a part of this group. In India only genetically modified cotton is under commercial cultivation since 2002. Lot of research is going on throughout the world including India on rice biotechnology. Rice is the first food crop for which complete genome sequence is available, which gives an opportunity to identify and functionally characterize the genes useful for different traits viz., agronomic performance, resistance to biotic and abiotic stresses and improved quality. There are two major approaches in using this technology for the crop improvement one is using the genes already present in other varieties or species of the same genus (Marker Assisted Selection) and the second one is through introgression of genes from other unrelated organisms (transgenics). The first one is just a modified method of traditional breeding except for using the new tools of selection whereas the second one is almost a new technology.

4.7.1. Somaclonal Variation

Selection of favourable mutant cells and somaclonal variant strains from callus culture is supplementary tools to traditional breeding for production of new crop genotypes. The introduction of a specific genotype in *in vitro* selection programme depends on potentiality of callus induction and embryogenic callus induction, subsequently green plantlet regeneration. Studies have shown that genotype affects *in vitro* culture response (Mandal and Roy, 2003, Roy and Mandal, 2005a; Roy *et al.*, 2011b). *In vitro* cell culture may be effectively used in genetic modification of crop plants. One approach of such genetic modification involves the isolation of stable variant cell lines from established cell culture either directly or following mutagen treatment.

In vitro selection of salt-tolerant cell lines and regenerated plants has been reported in several species, such as, potato (Sabbah and Tal, 1990), rice (Lutts *et al.*, 1999; Aditya and Baker, 2005), *Hordeum* (Sibi and Fakiri, 2000), wheat (Barakat and Abdel-Latif, 1996), sunflower (Alvarez *et al.*, 2003), sugarcane (Gandonou *et al.*, 2005).

4.7.2. Androclonal Variation

The F_1 another culture (AC) technique is one strategy for reducing evaluation process. AC shortens the breeding cycle, producing homozygous lines in one generation. In traditional breeding, breeder has to establish 3000-

5000 plants in F₂ generation alone, whereas, in AC only 60-70 plants are required (Khush and Senadhira, 1994). Snape (1989) reported on increase in selection efficiency using double haploid breeding compared to conventional practices. Thus, it should be easier to identify superior genotypes in a cross and produce new cultivars.

The AC method is based on gamete instead of sporophyte selection and the probability to obtain the described genotype is higher in haploid than diploids. The AC strategy should also enable a great use of minor genes for resistance, the cumulative effect of which may provide more desirable resistance than that provided by major genes (Toenniessen and Khush, 1991). The possibility of generating recombinant with desirable characteristics is high (Zapata *et al.*, 1991) in AC.

Effect of organic adjuvants, synthetic plant hormones and diverse carbon sources in influencing anther culture response in rice had been studied by Roy and Mandal (2005b). Androgenic callus induction as well as green plantlet regeneration was more when callus was induced on N6 fortified with 100 mg L⁻¹ YE. However, coconut water (5 and 10% concentrations from Pink Dwarf variety) encouraged green plantlet regeneration only in Karnal local 95.

4.7.2.1. Limitations in Development of In Vitro Culture

Although all the above cited experimental results described the tremendous potential of cell culture as a powerful technique for plant improvement, this approach has certain limitations and problems as:

1. Only a certain type of characters can be selected.
2. Till today there is no way of direct selection for increased yield through the cell culture techniques.
3. Metabolic adaptation by cultured cells, genotype-dependent characters, a dramatic decrease in the regeneration of embryogenic cultures (Lutts *et al.*, 1999) are characteristic features of anther culture.
4. Regeneration from long term callus culture is difficult and some time not possible. For instance, after 5 months of subculture, only 10% of subcultured callus differentiated into shoots (Ozias-Akins and Vasil, 1982).
5. The cultured cells frequently contain nuclei showing various degrees of polyploidy, aneuploidy and chromosome abnormalities. This chromosomal variation often leads to the loss of morphogenetic potentiality.
6. Usually many selected resistant clones become susceptible after passages on salt free medium.
7. Phenotypic changes in regenerated plants and non-expression of salt-tolerant phenotypes are frequently observed in regenerated plantlets

8. The difficulty of conserving other characteristics of the original varieties in plantlets selected for their tolerance to salt.

4.7.3. Marker Assisted Breeding

Plant breeders are presented with many possibilities for altering the behavior of existing crops to find a solution to the present day problems of yield stagnation, biotic and abiotic stress and quality. They may choose to use the marker assisted technology for facilitate the selection of favourable gene combinations that occur naturally within a species or via the introduction of alien genes using transgenics (Coffman, 2004).

Marker assisted breeding involves use of molecular markers to track the traits of interest Marker-assisted selection in a plant breeding context involves scoring indirectly for the presence, or absence, of a desired plant phenotype or phenotypic component based on the sequences or banding patterns of molecular markers located in or near the genes controlling the phenotype (Coffman, 2004). The banding pattern of the molecular marker at a given locus is indicative of the presence or absence of a specific gene or chromosomal segment that is known to carry the desired gene or allele. DNA markers located near the gene or allele co-segregate with the trait of interest across generations makes it easy to identify the presence or absence of particular gene/ allele in the lab itself without growing the crop. Many rice varieties were developed using this technique. IRRI developed the submergence tolerance varieties, Swarna Sub1, IR64 Sub1, Samba Mahsuri Sub1 by introgression of Sub1 locus from FR13A through marker assisted selection. In India, the bacterial blight resistant varieties were developed by introgression of bacterial blight resistant genes into Pusa Basmati 1 and Samba Mahsuri through marker assisted selection. IRRI developed near isogenic lines (NILs) of bacterial blight and blast resistant genes in the genetic background of IR 24 and Co 39 varieties.

4.7.4. Transgenic Development

Genetic engineering involves cloning and incorporation of one or more genes from other genera through transformation into rice. There are different methods of DNA transformation viz., using protoplasts (Datta *et al.*, 1990), biolistic guns (Christou *et al.*, 1991) and *Agrobacterium* (Hiei *et al.*, 1994) which are routinely used to transform rice (also see the review of Roy *et al.*, 2011b). The transgenics for different biotic and abiotic stresses were developed in different laboratories throughout the world. The vast majority of rural and urban poor are consuming the nutritionally deficient food. As a micronutrient induced malnutrition is highly prevalent in Asia. Developing micronutrient dense rice, with higher amounts of iron, zinc and vitamin A, can have tremendous impact on the health of the low income people (Underwood, 1999). Transgenic rice carrying agronomically important genes for resistance to biotic and abiotic stresses, nutritionally enrichment and tolerance to herbicide have been produced in both *japonica* and *indica* rice (Table 4.13).

Table 4.13. Some examples of transgenic rice plants with agronomically important genes

Transgene	Transfer method	Useful trait	Reference
<i>bar</i>	Microprojectile bombardment	Tolerance to herbicide	Cao <i>et al.</i> , 1992
<i>bar</i>	PEG-mediated	Tolerance to herbicide	Datta <i>et al.</i> , 1992
Coat protein	Electroporation	Tolerance to stripe virus	Hayakawa <i>et al.</i> , 1992
Coat protein	Particle bombardment	Tolerance to rice tungro spherical virus	Sivamani <i>et al.</i> , 1999
Chitinase	PEG-mediated	Sheath blight resistance	Lin <i>et al.</i> , 1995; Datta <i>et al.</i> , 2001
<i>cryIA(b)</i>	Electroporation	Tolerance to striped stem borer	Fujimoto <i>et al.</i> , 1993
<i>cryIA(b)</i>	Particle bombardment	Tolerance to yellow stem borer and striped stem borer	Wunn <i>et al.</i> , 1996; Ghareyazie <i>et al.</i> , 1997
<i>cryIA(c)</i>	Particle bombardment	Tolerance to yellow stem borer	Nayak <i>et al.</i> , 1997
<i>cry1A(b)</i> , <i>cryIA(c)</i>	<i>Agrobacterium</i> -mediated	Tolerance to striped stem borer and yellow stem borer	Cheng <i>et al.</i> , 1998
<i>cry1A(b)</i> , <i>cryIA(c)</i>	Particle bombardment	Tolerance to yellow stem borer	Tu <i>et al.</i> , 2000
<i>cry1A(c)</i> , <i>cry2A</i> , <i>gna</i>	Particle bombardment	Tolerance to stem borer, leaf folder and brown planthopper	Maqbool <i>et al.</i> , 1998, 2001
<i>CpTi</i>	PEG-mediated	Tolerance to striped stem borer and pink stem borer	Xu <i>et al.</i> , 1996
<i>gna</i>	Particle bombardment	Insecticidal activity for brown planthopper	Rao <i>et al.</i> , 1998
Corn cystatin (CC)	Electroporation	Insecticidal activity for <i>Sitophilus zeamais</i>	Irie <i>et al.</i> , 1996
<i>Xa21</i>	Particle bombardment	Resistance to bacterial blight	Tu <i>et al.</i> , 1998; Zhang <i>et al.</i> , 1998
<i>codA</i>	Electroporation	Increased tolerance to salt	Sakamoto and Murata, 1998
Soybean ferritin	<i>Agrobacterium</i> -mediated	Increased iron content in seed	Goto <i>et al.</i> , 1999; Lucca <i>et al.</i> , 2001
<i>psy</i>	Particle bombardment	Phytoene accumulation in rice endosperm	Burkhardt <i>et al.</i> , 1997
<i>psy</i> , <i>crt1</i> , <i>lcy</i>	<i>Agrobacterium</i> -mediated	Provitamin A	Ye <i>et al.</i> , 2000

Among the above transgenics only few have completed different steps of development and came up to field level whereas other are still in developmental phases. Golden rice and *Bt*-rice are few examples which are on the verge of release.

4.7.4.1. Golden Rice

The first golden rice was developed in the year 2000 in Taipei 309 a japonica variety background containing two transgenes (phytoene synthase and lycopene cyclase) from daffodil and one (phytoene desaturase) from *Erwinia bacterium* by the research team headed by Ingo Potrykus at the Swiss Federal Institute, Zurich and the German Scientist Peter Beyer (Ye *et al.*, 2000). The presence of these three genes (*psy*, *lyc*, *crt1*) ensures that a metabolite called geranyl geranyl-pyrophosphate, which normally accumulates in large quantities in rice endosperm, is being converted into *beta*-carotene, the precursor of vitamin A. The *beta*-carotene content in the first developed golden rice was low at 1.6 to 1.8 µg/g, but it proved the functionality of the genes in rice. With the bacterial gene and a change in the promoter of one gene from daffodil, a javanica variety Cocodrie was developed by Syngenta that contained 6 to 8 µg/g *beta*-carotene. This line was designated Golden Rice 1 and was donated by Syngenta in 2004 to the Golden Rice Humanitarian Board (<http://www.goldenrice.org>). In 2005, Golden Rice 2 was developed by Syngenta in the background of American long grain rice variety Kaybonnet (javanica rice), which contained maize and bacterial transgenes that produced up to 36.7 µg/g beta carotene which is four times higher compared to Golden rice 1 (Paine *et al.* 2005). Five countries viz., Philippines, India, Bangladesh, Vietnam, and Indonesia were identified as the countries where the GR2G event of golden rice 2 which produces up to 25 µg/g of beta carotene would be the only event to move forward through regulatory approvals and eventually released. It is expected that Golden Rice will be released in the Philippines and Bangladesh as early as 2012, followed by India, Indonesia and Vietnam.

In the Philippines one popular rice variety, PSB Rc-82 is being modified with the GR2G event by the Philippine Rice Research Institute (PhilRice) and in Bangladesh the GR2G event is being introgressed into important *Boro*-rice variety BR-29 by the Bangladesh Rice Research Institute (BRRI). In India 3 popular varieties, Swarna, MTU-1010 and ADT-43 are undergoing modification with GR2G at Indian Agricultural Research Institute (IARI), New Delhi, Directorate of Rice Research (DRR), Hyderabad and Tamil Nadu Agricultural University (TNAU), Coimbatore respectively in collaboration with Department of Biotechnology, New Delhi, University of Delhi, South campus, New Delhi.

4.7.4.2. Bt-rice

The genes encoding for toxins from soil bacterium, *Bacillus thuringiensis* (*Bt*) were first transferred in 1987 to tomato, tobacco and potato for protection

against lepidopteran insects pests. With the success of *Bt* Cotton it became well known in India and simultaneously its use in other crops has gained momentum. Fujimoto *et al.* (1993) introduced a truncated d-endotoxin gene, *cry1A(b)* into rice. Transgenic plants in the R₂ generation expressing the *cry1A(b)* protein showed increased resistance to striped stem borer and leaf folder. *Bt* rice has been field evaluated for several years (Tu *et al.*, 2000; Ye *et al.*, 2001) and has shown resistance against several insect pests viz., striped stem borer, yellow stem borer, pink stem borer, leaf folder and green semilooper. There is no proper resistant rice variety against the major insect pest yellow stem borer (*Scirpophaga incertulas*) which is wide spread in Asia. Therefore *Bt* rice which is resistant to this insect pest is a promising alternate in many countries including India, but its commercialization has not yet taken place anywhere in the world. In India, *Bt* rice have been developed by many Indian Council of Agricultural Research Institutes like Indian Agricultural Research Institute, New Delhi; Central Rice Research Institute, Cuttack etc. and State Agricultural Universities like Punjab Agricultural University, Ludhiana, Tamilnadu Agricultural Univesrsity, Coimbatore, Bose Institute, Kolkata etc.

4.7.4.3. Iron Rich Rice

Goto *et al.* (1999) inserted the coding sequence of the soybean *ferritin* gene into kita-ake, a rice cultivar through *Agrobacterium*-mediated transformation. The introduced ferritin gene was regulated by the rice seed storage protein glutelin promoter, *GluB-1*, and terminated by the *Nos* polyadenylation signal. The protein synthesis from this gene was confirmed through western blot analysis and its accumulation in endosperm through immunological testing. The iron content of transgenic rice (T1) was up to three times higher than the untransformed seeds.

4.8. SEED PRODUCTION

The seed production is almost similar to crop production mainly with respect to crop management and plant protection measures but additionally the seed quality parameters should be monitored during and after production. For seed production prior planning and scientific knowledge about the crop is essential.

4.8.1. Quality Parameters

The important quality parameters to be taken care of during seed production are- Genetic purity, Physical purity, Germination of harvested seed, Moisture content of harvested seed, Seed health.

4.8.1.1. Genetic Purity

Genetic purity is to be maintained both in the field and after harvest. To

maintain genetic purity at the field and seed level the following points are to be considered while taking up seed production.

1. **Source of the seed material:** The source seed should be of higher class than the class of seed being taken up for seed production. For breeder seed production source seed is nucleus seed where as for foundation seed production breeder seed and for certified seed production, foundation seed are source seed. The source seed should contain proper tag with quality details and is from an authorized producer.
2. **Land requirements:** The land to be used for seed production should be free from volunteer plants and should be puddled thoroughly. The previous crop should not be paddy, if it is paddy, it should be of same variety and same or higher class.
3. **Isolation requirements:** The minimum isolation distance for rice crop is 3 meters from fields of other varieties. This is due to occurrence of natural cross pollination up to 5% even in self pollinated crops.
4. **Roguing:** Removal of off-types or rogues or volunteer plants from the seed production plot is called rouging. In rice rouging should be done at least in three crop phases i.e., during active tillering stage, flowering stage and mature grain stage.
5. **Precautions during harvest and post harvest operations:** During harvesting, threshing, winnowing and processing proper care should be taken to avoid mechanical mixtures.

4.8.1.2. Physical Purity

The harvested seed should be free from physical impurities and from obnoxious weeds so that the minimum physical purity of 98% as per the standards can be achieved. In rice the wild rice is considered as the obnoxious weed. So removal of this weed should be done before harvesting the crop.

4.8.1.3. Germination

Since the final produce of seed production will be used for sowing of next crop it should have minimum germination percentage of 80% as per the standards i.e. at least 80 out of 100 seeds should be able to produce normal seedlings which can be established in the field.

4.8.1.4. Moisture Content

The harvested seed should be properly dried and contain maximum 13% of moisture as per the Indian minimum seed certification standards (Tunwar and Singh, 1988).

4.8.1.5. Seed Health

The seed crop should be free from diseased plants and even the seed should not contain pathogen infection. The diseased plants should be removed during rouging. Paddy bunt caused by *Neovossia horrida* is the designated disease in rice crop. The maximum limit of bunt infected seeds is 0.1% in foundation class and 0.5% in certified class by number.

4.8.2. Seed Rate

The amount of seed required per acre differs with the type of planting method. For normal transplanting 20-25 kg seed per acre is sufficient but for SRI method of transplanting only 2 kg seed is sufficient per acre of land. For direct seeding 16-20 kg seed is required for one acre area.

4.8.2.1. Direct Sowing

Direct sowing is of primitive method comparative to transplanting. But there are certain advantages like reduction in cost of production by reducing the seed rate, lack of nursery preparation; lack of transplanting etc. and the duration is shortened by a week or so. It is also having some disadvantages like weed control, susceptible to heavy downpour after sowing, needs perfect leveling etc. Direct seeding is of mainly two types i) Dry seeding method ii) Sprouted seeding method. In Dry seeding method the dry seed is sown either through broadcasting or through drum seeder. Drum seeder is useful to sow the seeds in lines and is advanced technology than the broadcasting method. For direct seeding only 16-20 kg of seed is required per acre.

4.8.2.2. Transplanting

Transplanting is considered to be evolved method compared to direct seeding. But it is also having both advantages and disadvantages. As said above the cost of production is comparatively high in case of transplanting but control of weeds, maintenance of optimum plant population, ease for intercultural operations etc. are some of the advantages in this method. Transplanting is of two types:

- i) Line transplanting
- ii) Jig-Jag or random method of transplanting.

In line transplanting the required plant population can be maintained compared to jig jag method. In transplanting during *Kharif* (rainy) season spacing of 20 × 15 cm is generally practiced. Row-to-row distance is 20 cm and 15 cm between plants-to-plant. During Rabi (winter) season 15 × 15 cm spacing is practiced. The reduction in the spacing during Rabi is mainly due to the availability of long sunshine hours along with increased intensity than *Kharif* season. For transplanting 20-25 Kg seed per acre is sufficient.

4.8.2.3. System Rice Intensification

Recently a new method of transplanting called System Rice Intensification (SRI) technology is getting popular because of its advantages over the traditional transplanting. SRI involves the use of certain management practices, which together provide better growing conditions for rice plants, particularly in the root zone, than those plants grown under traditional practices. SRI was first developed in Madagascar during 1980 and it has been spread to China, India, Indonesia, Philippines, Sri Lanka and Bangladesh.

There are six practices in SRI:

1. **Young seedlings:** Seedlings are planted early. Rice seedlings are transplanted when only the first two leaves have emerged from the initial tiller or stalk, usually when they are between 8-12 days old. While planting the seed sac should be kept attached to the infant root, because it is an important energy source for the young seedlings.
2. **Single seedling/hill:** Seedlings are planted singly rather than in clumps of two or three or more. This facilitates the individual plants to have more space to spread and to spread root system and do not compete with other plants for space, light and nutrients in the soil.
3. **Wider spacing:** Seedlings are planted in a square pattern with plenty of space between them in all directions. Usually they were planted with 25 ´ 25 cm spacing which helps for vigorous root growth and more tillering. This square pattern facilitates in situ incorporation of weeds by cono-weeder.
4. **Less seed rate:** SRI method requires much lower seed rate (5 kg/ha) than traditional methods (50-75 kg/ha).
5. **Moist but un-flooded soil conditions:** Rice has traditionally been grown in water under submerged conditions. In SRI method, soil is kept moist but not flooded during the growth period, ensuring that more oxygen is available in the soil for the roots, occasionally the soil should be allowed to dry to a point of hair thin cracking except in saline soils. This will allow oxygen to enter the soil and also induce the roots to grow and search for water.
6. **Cono-weeding:** Weeding can be done by cono-weeder. First weeding should be done on fourteen days after transplanting and this should be continued at 7-10 days interval afterwards. At least two or three weeding are required. This practice of churning of soil seems to improve soil structure and increases the aeration of the soil. Thus the incorporation of weed biomass into soil results in enrichment of CO₂ near root zone, increases the biological activities, increases soil microbes population and activities, results in better nutrient availability in soil and uptake by plants.

4.8.3. Seed Treatment

By nature itself rice grain is protected by the outer husk. The husk protects the inner embryo and endosperm from infections and infestations in general. But to prevent the seed borne and or seed transmitted diseases, seed treatment is the best available option. Seed treatment is generally done with carbendazim. For wet seed treatment one gram carbendazim is sufficient for one kg seed whereas for dry seed treatment 3g of carbendazim is required. Seed treatment with the help of seed drum gives uniform spread of the chemical on the seeds.

4.8.4. Seed Dormancy

The inability of the viable seeds to germinate even under favourable conditions is called *dormancy*. The level of dormancy varies among different varieties and ranges from zero dormancy (*viviparous germination*) to as long as seven weeks dormancy. For breaking the seed dormancy seeds should be treated with 0.63% (6.3 ml/lit of water) of conc. Nitric acid for 24 hours followed by washing and incubation for another 24 hours. One litre acid solution is sufficient for one kg of seed. The deeply dormant seeds (e.g. MTU 1001) should be treated with 1% conc. Nitric acid instead of 0.63%.

4.8.5. Nursery

The nursery area should have good irrigation and drainage facility. The land should be puddled two to three times at an interval of 5-7 days so as to minimize the percolation losses of water and nutrients. Separate drainage channels should be prepared for proper drainage. Soak the paddy seed for 24 hrs and incubate for another 24 hrs before sowing the sprouted seed in nursery. Five kg sprouted seed is sufficient per cent of nursery bed. Generally the nursery beds of two cents (80 m²) size are convenient for maintaining proper irrigation and to take up plant protection measures.

4.8.6. Land Preparation and Planting

Before transplanting the nursery in the main field, the land should be ploughed thoroughly followed by puddling at least ten days before transplanting after following perfect levelling with levelling planks. Perfect leveling aids in reducing the weed growth and facilitates better water management. The optimum plant density during rainy season (*Kharif*) is 33 hills/ sqm (20 × 15 cm) and 44 hills /sqm (15 × 15cm) during *rabi* season. The seedlings of 25 days age are best suitable for transplanting. Shallow planting with 2-3 seedlings per hill is desirable to have profuse tillering.

4.8.7. Weed Control

Weed growth should be checked during first six weeks after transplanting. Application of pre-emergence herbicides viz. Butachlor @ 1.25 kg /acre (or)

Pretilachlor @ 600 ml/acre or Anilophos @ 500 ml /acre or Oxadiargyl @ 35g/ ac by mixing 25 kg of sand is very effective in controlling the weeds in the initial stages of crop growth. These pre-emergence herbicides are to be applied after 3-5 days after transplanting to control grasses and sedges effectively. The broad leaved weeds can be controlled by applying 2,4-DEE granules @ 8-10 kg/acre after 3-5 days after transplanting. The post emergence herbicides viz. Pyrazosulfuron @ 80g/ac at 8-10 or Ethoxy sulfuron @ 50g/ac at 10-12 days after transplanting are effective against monocot weeds and 2,4-D Sodium salt (Fernoxone 80% WP) at 2 g/l of water at 20 days after transplanting is helpful in controlling broad leaved weeds. Maintenance of thin film of water at the time of herbicide application helps in effective control of weeds in the transplanted crop.

4.8.8. Fertilizer Recommendation

4.8.8.1. Nursery

In the Nursery basal fertilization with 0.5 kg of 'N'; 0.5 kg of 'P' and 0.5 Kg of 'K' per every 100 sq. m is required to get robust seedlings, followed by another 0.5 kg 'N' at 12 days after sowing. Whenever zinc deficiency is observed in the nursery, zinc sulphate @ 2.0 g /lit. should be sprayed once or twice at 5 days interval. In case of iron deficiency in the nursery, 5-10 g of Ferrous sulphate (or) Ferrous ammonium sulphate with 0.5 to 1.0 g of citric acid per litre of water should be sprayed.

4.8.8.2. Main Field

- Fertilizer should be applied as per the recommendations based on the soil test.
- Apply Farm Yard Manure @ 10 Tons per ha for improving the soil condition and health.
- Nitrogen is to be applied in three splits (at basal, at active tillering stage and at Panicle Initiation stage) Phosphorous and Potash may be applied as basal in heavy soils.
- In case of light soils, Potash may be applied in two equal splits i.e., one at basal and the other at panicle initiation stage.

4.8.8.3. Green Manuring

Green manuring is an important renewable source of input for building of soil fertility and supplementing plant nutrients, especially nitrogen. This practice is more appropriate for irrigated agriculture, due to easy mineralization of nutrients from the turned over plants in the soil. Green manures may be either leguminous crops grown *in situ* (*Sesbania* sps., *Crotalaria* sps.) or non

leguminous plants/ trees (*Gilicidia maculate*, *Pongamia*; *Calotropis* etc) grown on bunds of waste lands for utilizing vegetative parts for green leaf manuring of soils. In addition leguminous crops like cow pea, green gram and black gram are used as green manure or incorporation of vegetative parts after harvesting seed. Following are the advantages of green manuring:

- Leguminous green manure crops fix atmospheric nitrogen.
- It decomposes easily without leaving much residue in the soil.
- Green manures promote the activity of soil micro-organisms
- It improves soil water holding capacity
- It suppress weed growth in rice cultivated fields
- Rectifies micronutrient deficiencies
- It supplies nutrients especially nitrogen through addition of biomass

4.8.8.4. Micronutrient Deficiencies and Correction

a) Zinc deficiency

Symptoms

- Poor stand and establishment, lack of growth and tillering.
- Unhealthy and sickly appearance of crop at 3-4 weeks after transplantation even after receipt of adequate quantities of N,P and K fertilizers
- Basal mid rib bleaching of 3rd / 4th leaf from top at about 2-4 weeks after transplantation.
- Dark brown rusty spots on the upper part near the tips of older and mature leaves.

Causes

- The problem of deficiency is likely to aggravate due to very low night temperatures
- When large quantities of un-decomposed organic matter particularly paddy straw is applied at low temperatures
- In saline and alkaline soils
- Under ill drained conditions
- Excess use of N,P,K fertilizers
- In fields where top soil is removed
- Variety Vijetha is more prone to Zn deficiency

Correction

- Application of zinc sulphate @ 20 kg/ac at puddling is suggested as a prophylactic measure in Zinc deficient endemic soils
- Deficiency in the standing crop can be corrected by spraying zinc sulphate @ 0.2% (2 g/L of water). The spraying should be repeated at 5 days interval depending on the severity of the problem
- Zinc sulphate solution should not be mixed with any insecticide / fungicide and with phosphorous fertilizers.

b) Sulphide Injury**Symptoms**

- Sudden foul smell of hydrogen sulphide resembling that of rotten eggs emits from soil
- A black mark will be seen on the roots if plant is pulled out and examined, which turn to reddish brown when exposed to air.
- The crop starts wilting, older leaves become yellow and dry out prematurely
- Parched appearance of field will be observed.

Correction

- Application of nitrogenous fertilizer through neutral fertilizers like Calcium Ammonium Nitrate (CAN)
- Draining of soil frequently will reduce the sulphide injury.

4.8.9. Irrigation

- Avoid moisture stress at tillering, panicle initiation, flowering and grain filling stages.
- A thin film of water (2-3 cm) should be maintained at the time of weedicide application and should not be drained up to one week
- Maintain water level at 5 cm depth during first seven days after planting and thereafter 2cm depth up to completion of tillering.
- Maintain water level at 5 cm depth from panicle initiation to grain maturity.
- Drain the water before fertilizer application.
- Mid season drying discourages unproductive tillers.
- Drain the field one week before harvest.
- Drain the field and aerate whenever Sulphide injury occurs.

4.8.10. Plant Protection

4.8.10.1. Insect Pests

Following cultural practices recommended for reducing the build up of insect pests:

- Summer ploughing
- Growing of suitable resistant varieties especially in endemic areas
- Usage of recommended doses of fertilizers
- Clipping of the leaf tips of seedlings while planting
- Adopting normal spacing of 20 x 15 cm during kharif and 15 x 15 cm during rabi
- Formation of alleyways at every 2-2.5 m for proper aeration and for easy movement during pest and disease management operations.
- Alternate wetting and drying
- Weed management in the field as well as on the bunds.
- Judicious usage of chemicals based on the economic threshold levels of pests (Table 4.14).

Table 4.14. Economic threshold levels of important insect pests

Insect Pest	Stage of the Crop	Economic threshold level
Stem borer	Nursery and Tillering	One adult or one egg mass per one sq. m or 5% of dead hearts per sq. m.
	Panicle stage	2% of white ears per sq. m
Gall midge	Nursery and Tillering	One silver shoot per hill or 5% silvershoots per sq. m.
BPH/WBPH	Tillering	10-15 insects per hill
	After Flowering	20-25 insects per hill
Leaf folder	All stages	One to two damaged leaves per hill or 4% folded leaves per sq. m.
Hispa	Tillering stage	Two adults per hill or two damaged leaves per hill
Green leaf hopper	Nursery	One or two insects per sq. m
	Tillering	10 insects per hill
	Flowering	20 insects per hill
Gundhi bug	Flowering	One to two adults per hill
Climbing cut worm	All stages	4-5 larvae per sq. m.

i) Yellow Stem borer (Scirpophaga incertulas)

Yellow stem borer causes dead heart in young seedlings during tillering stage and white ears in panicle stage. The caterpillars enter the tiller to feed and cut the shoot at base producing dead hearts in young plants. After panicle initiation the damage results in chaffy dried panicles called white ears.

Chemical control

- In endemic areas seedling root dip treatment in 0.05% chlorpyrifos emulsion for one minute before transplanting should be adopted.
- Spray Cartap hydrochloride 50 WP 2.0 g or chlorantraniliprole @ 0.3ml or acephate @ 1.5 g or profenophos @ 2 ml per litre of water (or) apply cartap hydrochloride 4G @ 8 kg/acre or carbofuron 3G @10 Kg/ acre when the adult moths/egg masses @ one per sq.m are noticed in the field.
- Application of granules during panicle initiation stage will be very effective in controlling white ears.

ii) Gall midge (Orseolia oryzae)

The Gall fly or gall midge causes the silver shoot or onion leaf which is a tubular gall in place of normal shoot. The maggots enter the stem and feed on growing part results in the formation of shining tubular structure called silver shoot. The affected tiller does not produce any panicle.

Chemical control

- Apply phorate 10G @ 5 kg/acre or carbofuran 3G @ 10 kg/acre in 1 to 2 inches of standing water.
- Spray monocrotophos @ 1.6 ml/ litre water.

iii) Brown plant hopper (Nilaparvata lugens)

The Brown plant hopper causes hopper burn i.e. circular patches of drying and browning of plants. It also acts as vector to grassy stunt, ragged stunt and wilted stunt viruses. Both the nymphs and adults suck sap from the basal portion of the stem leading to drying and browning of plants. Since the insects moves in radial fashion it results in circular patches popularly known as hopper burns.

Chemical control

- Spray acephate @ 1.5 g or monocrotophos @ 2.2 ml or ethofenprox @ 2.0 ml or fenobucarb @ 2.0 ml or imidacloprid @ 0.25 ml or thiamethoxam @ 0.2 g or Buprofuzin @ 1.6 ml per litre of water.
- Spray fluid should be directed towards the base of the plant.

- Avoid spraying of combination of insecticides and synthetic pyrethroids.
- If second spray is warranted alternate the previous chemical preferably belonging to another group.

iv) Leaf folder (Cnaphalocrocis medinalis)

The leaf folder damage results in longitudinal folding of leaves, scrapping and drying of leaves. The larvae fold the leaves by fastening leaf margins with stitches and feed the green tissue by scrapping. This results in white membranous dried leaves.

Chemical control

- Spray cartaphydrochloride @ 2 g or acephate @ 1.5 g or profenophos @ 2 ml or chloripyriphos @ 2.5 ml or monocrotophos @ 1.6 ml or flubendamide @ 0.5 g per litre of water.

v) Hispa (Dicladispa armigera)

The Hispa damage results in white parallel streaks or white patches along the long axis of leaf. The grubs feed on the chlorophyll by mining into the leaves and the adults feed on chlorophyll between the veins of leaf lamina results in white parallel streaks or patches.

Chemical control

- Spray profenophos @ 2.0 ml or chlorpyriphos @ 2.5 ml or monocrotophos @ 1.6 ml per litre of water

vi) Green leaf hopper (Nephotettix virescens / N. nigropictus)

The damage results in yellowing/ browning of plants with stunted growth. It also acts as vector to rice tungro virus. Both nymphs and adults suck sap from the leaves. The infected leaves turn brown from tip downwards exhibiting hopper burn symptoms.

Chemical control

- Apply carbofuron 3G @ 10kg/acre or spray monocrotophos @ 2.2 ml or ethofenphos @ 1.5 or chlorpyriphos @ 2.5 ml per litre water.

vii) Rice gundhy bug (Leptocorisa oratorius)

The rice gundhy bug or earhead bug damage results in yellowing of leaf tip, chaffyness of individual grains and black spot on the grain. Obnoxious odour emanates on disturbing the bugs. Both nymphs and adults suck juice from developing grains during milky stage which results in chaffy grains. Black spots appear around the puncture made by the bugs in the grain.

Chemical control

- Spray monocrotophos @ 1.6 ml or apply carbaryl @ 10 kg/acre during evening hours.

viii) Climbing cut worm (Mythimna seperata)

The young larvae feed on foliage and the mature larvae cut off the ear heads or their rachis during nights. The panicles or grains are seen lying in the field.

Chemical control

- Irrigate the field and spray in the evening hours with any of the following combinations, dichlorvos @ 1.0 ml + endosulfan @ 2.0 ml (or) dichlorvos 1.0 ml + chlorpyrifos @ 2.5 ml per litre of water.

ix) Thrips (Stenchaetothrips biformis)

The damage symptoms contain yellow or silver streaks on leaves of young seedlings. Terminal rolling and drying of leaves from top to base are characteristics symptoms. Both nymphs and adults suck sap from emerging panicles resulting in chaffy grains.

Chemical control

- Spray monocrotophos @ 36 SL 1.6 ml or chlorpyrifos 20 EC @ 2.5 ml or phosphamidon 40 SL @ 2.0 ml/litre of water.

x) Panicle mite (Steneotorsonemus spinki)

Initially the damage occurs as black elongated spots on the leaf midrib followed by brown streaks on the innerside of leaf sheath. Later the damage spreads to panicle results in chaffyness of panicle and discolouration of grains.

- Spray profenophos 2.0 ml or dicofol 5.0 ml per litre of water

4.8.10.2. Diseases*i) Blast (Pyricularia grisea)*

The fungus effects leaf, node and neck of the panicle. On leaves the spots are typically spindle shaped with dark brown margin and grey centre. Coalescence of spots results in drying of leaves and burnt appearance. The neck region of the affected panicle becomes black, brittle, necrotic and break.

Management

- Dry seed treatment with Carbendazim 50 WP @ 3.0 g/kg seed or wet seed treatment with Carbendazim solution @ 1.0 gm/Litre of water / kg of seed for 24 hours before sowing.

- Adopt clean cultivation (keep the field bunds free from weeds).
- Application of limited nitrogenous fertilizers in more splits
- As soon as the disease noticed, one of the following chemicals may be sprayed
 - Tricyclazole 75 WP @ - 0.6 gm/litre of water
 - Isoprothiolene 40 EC @ - 1.5 ml/litre of water
- A need based second spray may be given at 10 days after the first spray.
- In neck blast endemic areas Tricyclazole 0.6 gm/L may be sprayed at advanced boot leaf stage.
- If needed after emergence of panicle Isoprothiolene@ 1.5 ml/L may be sprayed.

ii) Sheath blight (Rhizoctonia solani)

The infection results in black irregular elongated lesions on leaf sheath, near the water level. Coalescence of these spots results in drying of entire plant. Inside the culm and on leaf sheath large number of small spherical brown sclerotia are formed.

Management

- Dry seed treatment with Carbendazim 50 WP @ 3.0 g/kg seed or wet seed treatment with Carbendazim @ 1.0 gm/Litre of water / kg of seed for 24 hours before sowing
- Adopt clean cultivation (keep the field bunds free from weeds).
- Recommended dose of nitrogenous fertilizer should be applied in more number of splits.
- To control the disease an one of the following fungicides should be used
 - Hexaconazole 5 EC @ 2.0 ml/L
 - Validamycin 3L @ 2.0 ml/L
 - Propiconazole 25 EC @ 1.0 ml/L
- Second spray should be given at 15 days after the first spray if needed by altering the chemical.

iii) Red stripe disease

The cause is still not known.

Management

- Application of Carbendazim @ 1.0 gm/L as soon as the initial symptom is noticed helps in reducing the spread of the disease.

iv) Stem rot (*Sclerotium oryzae*)

The basal portion of the stem gets rotten and when the culm splits open internal blackening can be seen initially and in later stages the entire plant dies.

Control

- Use seed from disease free crop.
- Burn straw & stubbles in affected areas
- Avoid stagnation of water
- Drain out water completely once the appearance of the disease
- In endemic areas, spray validamycin or hexaconazole @ 2.0 ml /L or benomyl @ 1.0 g /L or carbendazim @ 1.0 g/L twice at 15 days interval after maximum tillering stage
- Summer ploughing helps in reducing the inoculums.

v) Sheath Rot (*Sarocladium oryzae*)

Leaf sheath enclosing the ear head exhibits dark brown or black circular to irregular patches. The panicle does not emerge fully from the flag leaf. White powdery growth of mycelium is seen inside the leaf sheath and also on the panicle. The panicle turns into dark brown colour with partially filled grains.

Management

- Apply recommended dose of fertilizers
- Adopt optimum spacing
- Spray carbendazim @ 0.5g or benomyl @ 0.5g or mancozeb @ 2.5 g per litre water.
- Soil application of gypsum in 2 equal splits of 200 kg/acre reduces the incidence.

vi) Brown leaf spot (*Helminthosporium oryzae*)

On leaves the rectangular to oval reddish brown spots with grey centre will appear. In later stages the spots will spread to leaf sheath and ear head also. On surface of grains brown lesions either partially or fully observed.

Management

- Field sanitation and removal of alternate hosts and infected debris
- Crop rotation
- Use of recommended dose of fertilizers
- Use of disease free seeds
- Seed treatment with 2-3 g mancozeb per kg of seed
- Spray mancozeb @ 2.5 g per litre of water.

vii) Narrow brown leaf spot (*Cercospora oryzae*)

Generally this disease appears in mature crop but sometimes in tillering stage also it is seen. It produces short linear brown spots on leaves, sheath and even on pedicels and glumes.

Management

- Spray carbendazim @ 1g or mancozeb @ 2.5g per litre of water

viii) Bacterial Blight (*Xanthomonas oryzae* pv. *Oryzae*)

Generally it is seen in later stages of crop. It is chiefly confined to leaves. Initially water soaked spots will appear at the tip of the leaf and spreads along the margin of leaf in an irregular wavy manner. The affected portion turns straw coloured.

Management

- Burn stubble and use optimum dose of fertilizers
- Avoid flooded conditions
- Use nitrogen in maximum splits
- Use tolerant / resistant varieties
- Spray streptomycin sulphate and tetracycline combination 300g + copper oxychloride 1.25 Kg per hectare.

ix) Rice Tungro disease (*Tungro virus*)

The affected plants get stunted and turn yellow to orange colour from base to top. The discolouration starts from tip to downwards. Older leaves exhibit rusty spots. The ear heads are small and grains are ill filled. The virus is transmitted by Green leaf hopper.

Management

- Grow tolerant / resistant varieties
- Control vector (GLH) using insecticides

- Destroy weed hosts
- Burn the infected plants after uprooting
- Burn the stubbles in infected field
- In endemic areas practice crop rotation

4.8.10.3. Rodent Control

For Endemic Areas

- Destruction of rodent harborage and observe rat moment.
- Reducing the number and size of field bunds
- Complete the sowing and planting uniformly in one area.
- From puddling to one month after planting, setup local traps @ 20 / acre.
- Installation of permanent bait stations from planting to flowering stages @ 5 /ha, four at corners of the field one meter inside the cropped area from the field bund and one at the centre. Bromadiolone bait @ 30 g per bait station should be replenished twice in a week.
- During crop period baiting with bromadiolone 0.005% baits prior to primordial initiation stage of the crop.
- From primordial initiation to crop harvest smoking of burrows with burrow fumigator is effective in reducing the rodent population.

Action Plan with Bromadiolone 0.005%

Day 1: Identify live burrows and simultaneously place 15 gm freshly prepared bromadiolone loose bait in packets inside the burrow.

Day 2: Repeat bromadiolone loose baiting in the active /live burrows.

Note: Control schedules should be exempted on community basis to check cross infestation through migration.

Unit requirement of Bromadiolone

- For 1st baiting to 50 burrows/ha @ 15 grams loose bait /burrow – 15 gm Bromadiolone required/ha
- For 2nd baiting to 10 burrows/ha @ 15 g loose bait per burrow (20 % of 50 burrows) – 3 g bromadiolone required /ha.

4.8.11. Harvesting and Storage

- Harvesting should be done when at least 80% of the grains are matured and turned to yellow. If the crop is harvested without proper maturity it leads to loss of viability of grains.

- Moisture percentage in the grain during harvesting should be less than 23%. For combine harvesting, moisture should not be greater than 18%.
- The harvested material should be dried in the field for 2-3 days.
- The grain should be free from inert material after threshing and winnowing.
- The winnowed grains should be sun dried on tarpaulin or pukka floor until the moisture content reaches 13%.
- Both over drying and under drying will lead to breakage of the grain during processing.
- High moisture content during storage leads to loss of viability due to increased grain respiration and attack of storage insects and pests.
- Paddy should be stacked in cool and dry places away from walls and on wooden or iron pallets.
- During storage regular monitoring of stored products is to be done to check insect infestation and germination%.
- Always mark the bag with the name of the variety to avoid chance mixing.
- The harvested produce should be tested at seed testing laboratory to check whether the harvested material meets the minimum certification standards (Table 4.15) or not for its usage as seed.

Table 4.15. Minimum seed certification standards in rice (Tunwar and Singh, 1988)

Character	Seed certification standards	
	Foundation	Certified
Pure seed (Min.)	98%	98%
Inert matter (Max.)	2%	2%
Huskless seeds (Max.)	2%	2%
Other crop seeds (Max.)	10/kg	20/kg
Other distinguishable varieties (Max.)	2/kg	2/kg
Total weed seeds (Max.)	10/kg	20/kg
Objectionable weed seeds (Max.)	2/kg	5/kg
Genetic purity (Min.)	99%	98%
Germination (Min.)	80%	80%
Moisture content (Max.)	13%	13%
For vapour proof containers (Max.)	8%	8%
Off types (Max.)	0.05%	0.2%
Tag colour	White	Azure blue

4.8.12. Seed Yield

The seed yield depends up on the variety yield potential, crop management, plant protection measures, soil fertility level and finally on the prevailing climatic conditions. In general, the high yielding varieties in fertile delta soils yields maximum up to 7.0 to 7.5 t/ha with an average of 5.0 to 6.0 t/ha. In rain fed up lands maximum seed yield is 3.0 to 3.5 t/ha and with an average of 2.0 t/ha.

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